

Fundamentals of Perpetual Futures*

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Abstract

Perpetual futures are the most popular cryptocurrency derivatives. Perpetuals offer leveraged exposure to their underlying without rollover or direct ownership. Unlike fixed-maturity futures, perpetuals are not guaranteed to converge to the spot price. To minimize the gap between perpetual and spot prices, long investors periodically pay shorts a funding rate proportional to this gap. We derive no-arbitrage prices for perpetual futures in frictionless markets and bounds in markets with trading costs. Empirically, deviations from these prices in crypto are larger than in traditional currency markets, comove across currencies, and diminish over time. An implied arbitrage strategy yields high Sharpe ratios.

JEL Classification: G11, G12, G13

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1 Introduction

Perpetual futures are, by far, the most popular derivative traded in cryptocurrency markets, generating a daily volume of more than \$100 billion. Prior to the recent collapse of FTX, perpetual futures were among the most actively traded products on the exchange, with the now-bankrupt hedge fund Alameda Research taking the other side of many such leveraged trades. Despite their central role in crypto markets, and their potential adoption by traditional derivative markets, there is relatively little work studying these derivatives. In this paper we ask: what are the theoretical fundamental values of perpetual futures, and how large are deviations from these fundamentals empirically?

Perpetuals are derivatives that allow investors to speculate on or hedge against cryptocurrency price fluctuations using high leverage, without needing to acquire the cryptocurrencies or to roll them over. Like traditional fixed-maturity futures, at initiation, a long and short counterparty agree on an initial futures price without exchanging any money, other than posting margin with the exchange. Both parties can enter or exit the contract at any time, with profits or losses continuously calculated and allocated to each side's margin account based on prevailing market prices.

Unlike fixed-maturity futures, perpetuals do not expire. This feature enhances the liquidity of the contract, as there is no staggering of contracts with different maturities traded on the exchange, and only a single perpetual futures contract per underlying asset is listed. Moreover, this instrument does not require market participants to roll over their futures positions and is traded 24/7.

Because they have no set expiration date, perpetuals are not guaranteed to converge to the spot price of their underlying asset at any given time, and the usual no-arbitrage prices for perpetuals are not applicable without additional restrictions. To minimize the gap between perpetual futures and spot prices, long position holders periodically pay short position holders a 'funding rate' proportional to this gap to incentivize trades that narrow

it. For example, when the futures price exceeds the spot price, arbitrageurs who borrow cash to long the spot and simultaneously short the futures would receive the funding rate. Their trades would tend to increase the spot price and decrease the futures price. A narrow gap means that perpetual futures offer effective exposure to variation in the spot price of the underlying asset to hedging and speculating investors. Typically, the funding rate is paid every eight hours and approximately equals the average futures-spot spread over the preceding eight hours. Note that the strategy just sketched, commonly referred to as 'funding rate arbitrage,' is not risk-free even disregarding margin requirements and trading costs, simply because there is no predetermined expiration date when the trade would be unwound at a profit.

We derive no-arbitrage prices for perpetual futures in frictionless markets and derive no-arbitrage bounds in markets with trading costs. The theoretical perpetual futures price is proportional to the spot price of the underlying, with a constant of proportionality determined by the interaction of interest rates and the exchange's funding rate mechanism. A key feature of this mechanism is the 'clamp' - when futures and spot prices are close, the funding rate is fixed at a small positive value, but when they diverge significantly, the funding rate increases linearly with the price gap to incentivize convergence. The interest rate captures the cost of borrowing cash to finance holding the underlying, while the funding rate intensity captures the benefit of shorting the futures. Thus, intuitively, the future-spot spread is larger when the cash borrowing interest rate is large relative to the funding rate.

Our derivation relies on a weaker notion of arbitrage that we call *random-maturity arbitrage*. As its name suggests, unlike traditional riskless arbitrage, we allow the strategy's time-to-maturity to be random. At first glance, one might object that such prices are not truly based on riskless arbitrage. Note, however, that riskless no-arbitrage pricing is usually just a useful fiction ([Pedersen, 2019](#)). For example, in real-world futures markets, arbitrageurs must maintain a margin account during the entire period in which

the arbitrage trade is open. Temporary worsening of apparent arbitrage opportunities can lead to liquidations and losses. As the saying goes, an arbitrageur must remain liquid longer than the market stays irrational. Thus, even arbitrage opportunities that appear to be riskless in theory, may be risky in practice.

Our no-arbitrage prices provide a useful benchmark for perpetual futures and simultaneously prescribe a strategy to exploit divergence from these fundamental values. Motivated by the theoretical understanding, we study the empirical deviations of the perpetual futures price from the spot. We find that the mean deviation from our benchmark is modest and statistically insignificant, which suggests that it is a useful benchmark for futures prices.

Around this benchmark, however, perpetual futures often differ significantly from spot prices. The mean *absolute* deviation is about 52% to 90% per year across different cryptocurrencies, which is considerably larger than the deviations documented in traditional currency markets by [Du, Tepper, and Verdelhan \(2018\)](#). We find strong comovement of the futures-spot gap across different cryptocurrencies. This comovement can be due to commonality in funding and market liquidity faced by arbitrageurs who operate in multiple cryptocurrencies. Common sentiment could also drive the difference in futures demand relative to the spot. Overall, the deviation's magnitude is comparable to our theoretic no-arbitrage bound calibrated to actual trading fees.

Deviations decline on average about 22% a year, consistent with an increase in arbitrage capital and competition among arbitrageurs ([Kondor, 2009](#)). The narrowing gap provides an additional perspective on the downfall of Alameda Research and Three Arrows Capital. According to news reports and interviews, both hedge funds seem to have pivoted from such arbitrage activity around late 2021 to early 2022 and started taking more directional bets on cryptocurrencies, with both direct unhedged crypto holdings and investments in crypto startups. The large declines in crypto prices in 2022 subsequently exhausted their

capital and led to their bankruptcies.¹

To understand the economics of the futures-spot spread, we consider a trading strategy motivated by the random-maturity arbitrage theory. Whenever the futures-spot spread exceeds the theoretical bound for a given trading cost tier, we open the trading position and close it when the futures-spot spread returns to its theoretical relationship under no trading costs. We find that empirically, the random maturity arbitrage strategy generates a sizable Sharpe ratio even under high trading costs. For example, for Bitcoin perpetual futures, the strategy generates a Sharpe ratio of 3.35 under high trading costs typical of retail investors, and up to 11.65 for highly-active market makers who pay no such fees. The strategy's performance is even better for Ether and other cryptocurrencies and delivers significant alphas relative to the 3-factor model of [Liu, Tsyvinski, and Wu \(2022\)](#) and the 5-factor model of [Cong, Karolyi, Tang, and Zhao \(2022b\)](#).

The strong performance of our simple trading strategy, which exploits deviations from the random-maturity no-arbitrage bounds we derive, provides compelling evidence for the usefulness of these bounds as a benchmark for perpetual futures prices. The strategy generates high Sharpe ratios, even for investors subject to the highest trading costs on Binance. This indicates that the no-arbitrage prices serve as an effective anchor for perpetual futures, with profitable random maturity arbitrage opportunities arising when market prices diverge significantly from these fundamental values. Thus, the success of the trading strategy underscores the practical value and validity of our theoretical no-arbitrage results.

What explains these large no-arbitrage deviations? One natural explanation is that liquidity in crypto markets is insufficient for arbitrageurs to eliminate such violations. Our finding that the spreads decline over time is consistent with liquidity improving as these

¹See, for example, Forbes, November 19, 2022, on [How Did Sam Bankman-Fried's Alameda Research Lose So Much Money?](#), Odd Lots, November 17, 2022, on [Understanding the Collapse of Sam Bankman-Fried's Crypto Empire](#), and Hugh Hendry's interview on December 3, 2022 of Kyle Davies on the [Collapse of Three Arrows Capital](#).

markets develop and leaves open the possibility that they will narrow going forward. We also find, however, that past return momentum significantly explains the futures-spot gap with a time-series regression R^2 of more than 15%. When past returns are high, futures tend to trade at a higher price relative to the spot. This indicates positive feedback or momentum trading behavior in the perpetual futures market. This correlation may linger even as crypto markets become more efficient.

The existing literature on perpetual futures mainly focuses on empirical evidence. See e.g. [Alexander, Choi, Park, and Sohn \(2020\)](#), [De Blasis and Webb \(2022\)](#), [Ferko, Moin, Onur, and Penick \(2022\)](#), and [Streltsov and Ruan \(2022\)](#). [Christin, Routledge, Soska, and Zetlin-Jones \(2022\)](#) present a novel study on the strategy of longing the spot and shorting the perpetual futures to understand the demand pressure in the market. They document a significant return to this trading strategy and attribute the profits to differences of opinion and leverage constraints in the market. We find that our dynamic real-time trading strategy outperforms their directional bet even under a high trading cost scenario. Our paper provides a general explanation for their findings as perpetual futures are more expensive relative to the theoretical no-arbitrage price for most of the recent sample period across major cryptocurrencies. Therefore, longing the spot and shorting the futures would be consistent with arbitrage trading directions.

On the theory side, [Angeris, Chitra, Evans, and Lorig \(2022\)](#) provide a theoretical no-arbitrage analysis of perpetuals, which relies on a strong assumption that the payoff from the perpetual is a fixed function of the underlying spot price and focuses on the security design problem. Their approach would not be able to analyze arbitrage trading in case of a deviation from the no-arbitrage price. [Ackerer et al. \(2023\)](#) derive no-arbitrage prices for various perpetual contracts in both discrete and continuous-time by assuming the existence of an equivalent martingale measure. Our work advances this literature in three key ways: First, we derive no-arbitrage prices that incorporate realistic funding specifications, including the clamp feature used by major exchanges like Binance. Second,

we provide explicit arbitrage bounds and trading strategies for markets with transaction costs. Third, we demonstrate empirically that our benchmark improves arbitrage strategy performance compared to existing approaches. We further relate our work to both papers below.

Also related is a recent literature on fixed maturity futures in the crypto market. [Schmeling, Schrimpf, and Todorov \(2022\)](#) provide a comprehensive analysis of the carry of fixed-maturity crypto futures, with the carry defined following the general definition of [Kojien, Moskowitz, Pedersen, and Vrugt \(2018\)](#). They document volatile convenience yields for fixed-maturity crypto futures and attribute this to trend-chasing small investors and the relative scarcity of arbitrage capital. [Cong, He, and Tang \(2022a\)](#) provide a novel link of the volatile convenience yield to the staking, service flow, and transaction convenience of the underlying tokens. They show that the large deviation from uncovered interest rate parity can be reconciled with transaction convenience. Our paper focuses on perpetual futures rather than fixed maturity futures and extends fixed-maturity to random-maturity no-arbitrage pricing.

More broadly, our paper contributes to the understanding of frictions and arbitrage in cryptocurrency markets. [Makarov and Schoar \(2020\)](#) study price deviations across exchanges. They find large gaps across countries, highlighting the important role played by capital controls and slow-moving arbitrage capital as in [Duffie \(2010\)](#). Our analysis focuses on the price wedge between the spot and the futures market. We find that even within an exchange, futures prices deviate from their theoretical arbitrage-free values. These results indicate there are significant limits to arbitrage as in [Gromb and Vayanos \(2018\)](#) for cryptocurrencies in the early years. Finally, our finding that deviations from the no-arbitrage benchmark in crypto markets are highly correlated across currencies suggests that a common risk factor drives this variation as documented by [Lustig, Roussanov, and Verdelhan \(2011\)](#) in traditional currency markets.

The rest of the paper is organized as follows: Section 2 provides the institutional details and history of perpetual futures. Section 3 presents the no-arbitrage analysis of the perpetual futures market and derives the theoretical price of perpetual futures. Section 4 demonstrates the empirical futures-spot deviation and presents the simple theory-motivated trading strategy that can exploit the arbitrage opportunity. Section 5 provides some explanation for the deviation between futures and the spot. Section 6 concludes.

2 An Introduction to Perpetual Futures

The idea of perpetual futures was first introduced by Shiller (1993). The goal was to set up a perpetual claim on the cash flows of an illiquid asset. For example, the underlying illiquid asset could be a house that generates rents as cash flows. The purpose of the perpetual futures is to enable price discovery for the underlying with an illiquid or hard-to-measure price. Perpetual futures have no expiration date but cash is exchanged between the long and the short side: after buying the perpetual futures, the long side is entitled to receive the flow cash flow from the short side and they settle the price difference when exiting the position.

Perpetual futures in crypto markets similarly have no expiration date and cash is exchanged between the long and the short side, but their purpose is different from Shiller’s original idea. First, unlike, e.g. real estate market, crypto has no inherent dividend or cash flow. Second, the price discovery argument of Shiller is most applicable to settings where spot prices are difficult to measure. Crypto spot prices, however, can be measured from active trading on different exchanges, and decentralized exchanges such as Bancor (Hertzog et al., 2017) or Uniswap (Adams et al., 2021) offer price discovery for assets with minimal liquidity. The major role perpetual futures play in the crypto space is to offer an effective leveraged trading vehicle to hedge or speculate the underlying spot price movement, which makes the market more complete. It also serves as an effective tax

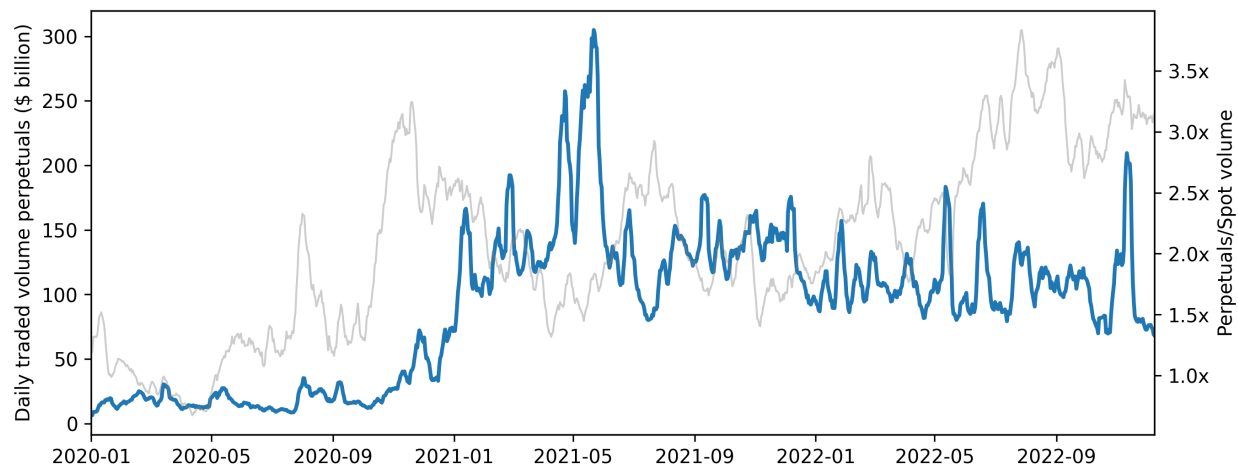


Figure 1 Total trading volumes of perpetual futures across exchanges

The figure displays the 7-day moving average daily traded volume for perpetual futures across all exchanges in blue. The median daily volume is \$17.8 bn. (2020), \$132.0 bn. (2021) and \$101.9 bn. (2022). This translates to a yearly volume of \$8,551 bn. (2020), \$51,989 bn. (2021) and \$39,306 bn. (2022) respectively. Additionally, the grey line depicts the ratio between the traded volume in perpetuals and spot markets. In 2022 the Perpetual markets are consistently trading between 2x or 3.5x the spot volume. The data is obtained from CoinGecko, a crypto data specialist. We exclude exchanges that are known for misrepresenting data (e.g. forms of wash trading).

payment optimization tool for investors.

Crypto perpetual futures were first introduced by BitMEX in 2016, which has gained great popularity in the crypto space since its inception. It initially served as an effective hedging tool for crypto miners. It was later adopted by crypto speculators interested in leveraged exposure. Nowadays, based on data from CoinGecko, the median total daily trading volume of perpetual futures across all exchanges is 101.9 billion in the year 2022 which is about $2\times$ to $3\times$ the total spot trading volume across these exchanges. Figure 1 presents the 7-day moving average of the total trading volume of perpetual futures across all exchanges. We see a significant rise in trading of perpetual futures around January 2021 and the total volume stabilizes at a level above \$100 billion per day following the rise.

Cong, Li, Tang, and Yang (2022c) document significant wash trading behavior among crypto exchanges because of competition, the ranking mechanism, and lack of regulation.

They estimate that over 70% of crypto trading volume is not real. [Amiram, Lyandres, and Rabetti \(2021\)](#) confirm this conclusion using new data and an extended methodology. Therefore, in calculating trading volume, we exclude exchanges that are known to misrepresent data.

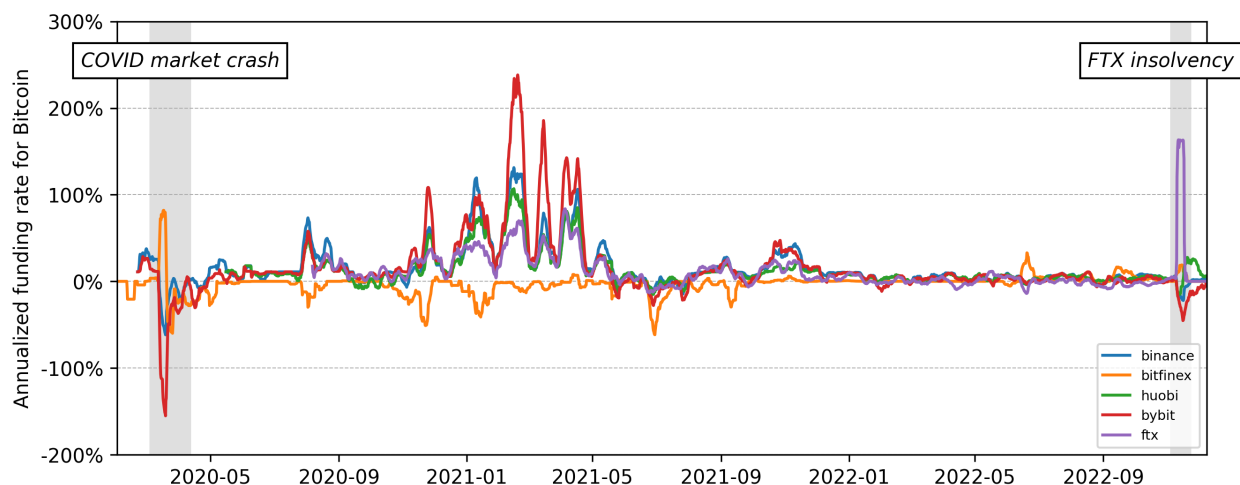


Figure 2 Bitcoin annualized funding rate across exchanges

The figure presents the 7-day moving average annualized funding rate for Bitcoin (BTC) across exchanges. The data is obtained from Glassnode, an analytics platform. The data covers two major market turbulences: the COVID stock market crash from February to April 2020 and the FTX insolvency in November 2022. The FTX collapse led to significant negative funding rates on all solvent exchanges. Thus, the perpetual future prices were lower than the spot prices on the solvent exchanges. Vice versa on the insolvent FTX. The funding rates become more volatile after the event, indicating increased uncertainty for the market participants.

A key feature of crypto perpetual futures is the funding rate, which is the cash exchanged between the long and short counterparties. Its goal is to keep the futures price close to the underlying spot so that the futures can be an effective hedging tool for spot price movement. The funding rate is typically paid every 8 hours. Its value is approximately a weighted average of the prior 8 hours' price gap of the futures and the spot. If the futures price is above the spot, the funding rate will be positive, meaning the long side of the futures needs to pay the short side. This incentivizes traders to short the futures, and in doing so, to move its price back in line with the spot. On the other hand, when the futures price is below the spot, the funding rate turns negative, which means the

short side needs to pay the long side. Therefore, the funding rate is the key mechanism keeping the futures price close to the spot price.² Note that perpetual futures cannot be replicated by rolling over 8-hour maturity futures. For the latter, every 8 hours, the price is guaranteed to converge to the underlying spot, while for the perpetual futures this is not the case.

Figure 2 presents the annualized 7-day moving average of Bitcoin perpetual funding rates across leading exchanges from January 2020 to December 2022. The funding rate is positive when the futures-to-spot spread is positive, and negative otherwise. Funding rates tend to be similar across exchanges due to cross-exchange arbitrage activity, but can diverge during extreme liquidity episodes. During March 2020, as COVID-19 started to spread and liquidity evaporated, funding rates turned substantially negative in most exchanges. Funding rates turned highly positive during the crypto bull run of early 2021. The last episode highlighted is the collapse of FTX, which was the 4th largest crypto exchange at the time. The figure shows that Bitcoin futures prices were substantially higher than spot prices at FTX, but the opposite was true on other exchanges. This pattern is consistent with FTX investors liquidating their short futures positions quickly, either voluntarily to reduce their exposure to the failing exchange, or involuntarily as the exchange liquidated their underfunded positions.

3 Arbitrage in Perpetual Futures

We next derive no-arbitrage prices for perpetual futures prices relative to spot prices. Unlike traditional futures, perpetual futures have no expiration date. To analyze arbitrage in this market, we extend the traditional notion of risk-free arbitrage, where arbitrageurs have a guaranteed positive payoff at a certain time in the future. We first describe the

²<https://www.binance.com/en/support/faq/360033525031> provides a detailed explanation of how the funding rate is calculated. We summarize the mechanism in Appendix B.

payoff structure of perpetual futures. We then introduce a generalized notion of an arbitrage opportunity with a certain positive payoff but at an uncertain future time.

Definition 1 (Perpetual futures). *A perpetual futures contract with price $\{F_t\}_{t=0}^\infty$ written on an underlying asset with spot price $\{S_t\}_{t=0}^\infty$ is a swap agreement between the long and the short side. There is no cost to open the position. Once open, both the long side and the short side can terminate their position at any time t . To stay in the contract, both sides have to meet two requirements. (1) For each unit of the perpetual, the long and short sides must exchange an $\mathcal{F}_s$ -adapted cash-flow $\Phi F_s ds$, $s \in (0, t)$, referred to as the funding payment. The funding rate function $\Phi : \mathbb{R} \rightarrow \mathbb{R}$ is a univariate function of the premium index $\frac{F-S}{F}$. If Φ is positive, the long pays the short, and vice versa. (2) Both sides need to post mark-to-market margins to cover any losses in their margin accounts.*

The traditional notions of arbitrage typically consider a guaranteed positive payoff at a certain future date. For example, [Cochrane \(2009\)](#) defines arbitrage as follows.

Definition 2 (Riskless arbitrage). *A riskless arbitrage opportunity is defined with respect to a payoff x at a certain future time τ and its price $p(x)$. If the following conditions are satisfied: (1) $x \geq 0$ almost surely, (2) $x > 0$ with some positive probability, and (3) its price satisfies $p(x) \leq 0$, then this payoff is an arbitrage opportunity.*

There is no certain future time, however, when a perpetual futures expires and its price is guaranteed to converge to some fixed function of the spot price. Therefore, a generalized notion of arbitrage is required. Suppose the risk-free rate is constant. We define *random-maturity arbitrage* opportunities as zero-cost strategies with a guaranteed positive payoff at a bounded uncertain future time. Stated formally:

Definition 3 (Random-maturity arbitrage). *A random-maturity arbitrage opportunity is defined with respect to a random payoff x at a bounded future random time $\tilde{\tau} < \bar{T}$, and its price $p(x)$. If the following conditions are satisfied: (1) $x \geq 0$ almost surely, (2) $x > 0$ with some positive*

probability, and (3) its price satisfies $p(x) \leq 0$, then this payoff is a random-maturity arbitrage opportunity.

Definition 3 generalizes traditional arbitrage in the sense that there is a guaranteed positive payoff but at an uncertain future time. The following corollary specializes this definition for perpetual futures:

Corollary 1. *In the perpetual futures market, if a strategy (1) has 0 cost at time 0, and (2) for any price path of the futures and the spot, $\{F_t\}_{t=0}^\infty$ and $\{S_t\}_{t=0}^\infty$, there exists a bounded unwinding time $\tilde{\tau}$ such that the strategy's discounted payoff is positive, then this strategy is a random-maturity arbitrage.*

We next show that when there is no random-maturity arbitrage, the futures price is a fixed constant times the spot price. First, we define some mathematical notation and state three key assumptions needed for our results.

For a function f , we define its generalized (set-valued) inverse as: $f^{-1}(y) = \{x : f(x) = y\}$. Our analysis relies on the following assumptions:

Assumption 1. *The risk-free short rate r in the cash market is constant.*

Assumption 2. *The gap between the perpetual futures and its fundamental price is bounded: $\sup_t |F_t(\omega) - \lambda S_t(\omega)| < \infty, \forall \omega$, where λ is a proportionality constant implied by Equation (1).*

Assumption 3. *The funding rate function $\Phi : \mathbb{R} \rightarrow \mathbb{R}$ is a univariate function of the premium index $\frac{F-S}{F}$. The function Φ is weakly increasing in its argument and $\sup\{x : \Phi(x) = r\} < 1$.*

Assumption 1 guarantees that there is no roll-over risk for the arbitrageurs. Assumption 2 is a no-bubble condition, which says that the largest deviation of perpetual futures from its no-arbitrage price can be bounded. Assumption 3 states that the design of the contract needs to guarantee that there is a reward for convergence trade. The reward should be weakly increasing when the deviation becomes larger. The last definition is

a technical condition that ensures the equilibrium futures price is positive. With these three assumptions, we can state our result as follows.

Proposition 1 (No-arbitrage price: General Case). *Suppose $\Phi^{-1}(r)$ is a single point, under Assumptions 1, 2, and 3, there is an absence of random-maturity arbitrage opportunities if and only if*

$$F_t = (1 - \Phi^{-1}(r))^{-1} S_t. \quad (1)$$

Proof. In this proof, we first demonstrate how the benchmark price in Equation (1) is derived. Next, we establish that if the futures price satisfies this relationship, random-maturity arbitrage opportunities are absent. Finally, we prove that the absence of random-maturity arbitrage opportunities necessitates that the futures price conforms to Equation (1), as any deviation would violate this assumption.

To show how we obtain the formula, start off with the conjecture that in equilibrium, the futures price will be a linear function of the spot, $F_t = \lambda S_t$, where λ is the proportionality constant in Equation (1).

If this conjecture is true, it implies both strategies laid out in Table 1 should always have 0 payoff. We can write down the payoff of the first strategy as:

$$-\int_0^t e^{-rs} d(F_s - \lambda S_s) + \int_0^t e^{-rs} (\Phi F_s - r \lambda S_s) ds = \int_0^t e^{-rs} (\Phi - r) F_s ds = 0, \quad (2)$$

where the first equality follows from the condition $F_t = \lambda S_t, \forall t$. From the second equality, we obtain the key condition for pinning down the value of λ :

$$\Phi = r, \quad (3)$$

which states that in equilibrium, the funding rate must equal the risk-free rates. When

$\Phi^{-1}(r)$ can be mapped to a single point, it pins down the equilibrium premium index:

$$\frac{F - S}{F} = \Phi^{-1}(r). \quad (4)$$

Rearranging the terms from the above equation gives us exactly the relationship between F and S as given in Equation (1).

		$F_0 > \lambda S_0$	$F_0 < \lambda S_0$
Actions		Long λ spot, short 1 futures	Long 1 futures, short λ spot
Time 0	Futures	0	0
	Spot	$-\lambda S_0$	$+\lambda S_0$
	Cash	$+\lambda S_0$	$-\lambda S_0$
Time t	Futures	$-\int_0^t e^{-rs} dF_s$	$\int_0^t e^{-rs} dF_s$
	Funding	$\int_0^t \Phi F_s e^{-rs} ds$	$-\int_0^t \Phi F_s e^{-rs} ds$
	Spot	$\lambda \int_0^t e^{-rs} dS_s$	$-\lambda \int_0^t e^{-rs} dS_s$
	Cash	$-\lambda \int_0^t e^{-rs} r S_s ds$	$\lambda \int_0^t e^{-rs} r S_s ds$
Payoff		$-\int_0^t e^{-rs} d(F_s - \lambda S_s) + \int_0^t e^{-rs} (\Phi F_s - r \lambda S_s) ds$	$\int_0^t e^{-rs} d(F_s - \lambda S_s) - \int_0^t e^{-rs} (\Phi F_s - r \lambda S_s) ds$

Table 1 Discounted payoffs to arbitrage strategies in perpetual futures and spot markets

This table presents the costs and benefits of two arbitrage trading strategies: (1) long λ spot and short 1 futures; (2) long 1 futures and short λ spot. In the last row, the payoff from exiting the position is the discounted payoff from futures and spot price changes, proceeds from the cash market, and the funding rate.

If Equation (1) always holds, then both strategies in Table 1 have zero payoff forever. Any other long-short strategy with different proportionality would be exposed to directional risk in the spot market. Therefore, this implies that there is an absence of random-maturity arbitrage opportunity across the futures and spot markets.

Next, we demonstrate that any deviation from this benchmark results in random-maturity arbitrage opportunities. We are going to show that if there is no arbitrage, then

$$F_t = (1 - \Phi^{-1}(r))^{-1} S_t. \quad (5)$$

Before proceeding, we define the deviation from the fundamental price as:

$$u_s \equiv F_s - (1 - \Phi^{-1}(r))^{-1} S_s = F_s - \lambda S_s. \quad (6)$$

Scenario 1: Firstly, consider the scenario where $F_0 > \lambda S_0$. In this scenario, we consider the first strategy in Table 1. Its payoff can be written as:

$$-\int_0^t e^{-rs} du_s + \int_0^t e^{-rs} (\Phi F_s - r \lambda S_s) ds = -e^{-rt} u_t + u_0 + \int_0^t e^{-rs} (\Phi - r) F_s ds, \quad (7)$$

where the equality follows from integration by parts. To prevent the arbitrageur from making a profit, the payoff must be nonpositive for all times t :

$$\underbrace{u_0}_{\text{initial spread}} + \underbrace{\int_0^t e^{-rs} (\Phi - r) F_s ds}_{\text{funding payments}} \leq \underbrace{e^{-rt} u_t}_{\text{spread at unwinding}}. \quad (8)$$

Consider the first time period when the deviation is positive and let t^* denote the first time it hits zero again. That is, $u_s > 0, s \in (0, \min\{t^* > 0, u_{t^*} = 0\})$.

If $t^* < \infty$, then when $t = t^*$, the RHS of Inequality (8) equals 0. On the LHS, when $u_s > 0$, $F_s > \lambda S_s \implies \Phi(\frac{F_s - S_s}{F_s}) > r$ because of the increasing property of Φ . Therefore, the LHS of Inequality (8) is larger than 0. Therefore, the payoff at t^* to the arbitrageur is positive. It remains to show that if $t^* \rightarrow \infty$, the payoff to the arbitrageur will also be positive.

Let $\underline{u}_t$ denote a process, implicitly defined by $e^{-rt} \underline{u}_t = u_0 + \int_0^t (\Phi - r) F_s e^{-rs} ds$, which provides a lower bound for all processes u_t satisfying Inequality (8). From the Equation, the deviation needs to increase without bound to keep the arbitrageur from making a profit, which would violate Assumption 2.

Therefore, if $F_0 > \lambda S_0$, the proposed strategy is a random maturity arbitrage strategy.

Scenario 2: Next, consider the scenario where $F_0 < \lambda S_0$. In this scenario, consider the

second strategy in Table 1. The payoff to this strategy is given by:

$$\int_0^t e^{-rs} du_s - \int_0^t e^{-rs} (\Phi F_s - r\lambda S_s) ds = e^{-rt} u_t - u_0 - \int_0^t e^{-rs} (\Phi - r) F_s ds. \quad (9)$$

In order to keep the arbitrageur away from making a profit, the above expression should satisfy:

$$e^{-rt} u_t \leq u_0 + \int_0^t e^{-rs} (\Phi - r) F_s ds. \quad (10)$$

Consider again the first time period the deviation is negative and let t^* denote the first time it hits zero again. That is $u_s < 0, s \in (0, \min\{t^* > 0, u_{t^*} = 0\})$.

If $t^* < \infty$, then when $t = t^*$, the LHS of Inequality (10) is 0. On the RHS, when $u_s < 0$, $F_s < \lambda S_s \implies \Phi(\frac{F_s - S_s}{F_s}) < r$ because of the increasing property of Φ . Therefore, the RHS of Inequality (10) is negative. Therefore, the payoff to the arbitrageur at time t^* is positive. It remains to show that if $t^* \rightarrow \infty$, the payoff to the arbitrageur will also be positive.

Let $\bar{u}_t$ denote a process, implicitly defined by $e^{-rt} \bar{u}_t = u_0 + \int_0^t e^{-rs} (r - \Phi) F_s ds$, which provides an upper bound for all processes u_t satisfying Inequality (10). From the Equation, the deviation needs to decrease without bound to keep the arbitrageur from making a profit, which would violate Assumption 2.

Therefore, if $F_0 < \lambda S_0$, longing 1 futures and shorting λ spot is a random maturity arbitrage strategy. $\square$

To gain an intuition for this result, consider the three terms of Equation (8). The first term is the initial spread. The second term is the present value of cumulated funding payments from initiation to unwinding. These first two terms account for gains to the arbitrageur in this scenario. The last term is the spread at unwinding discounted to the present, which is the source of tension here. For fixed-maturity futures, this spread is guaranteed to be zero at expiration. However, an arbitrageur in perpetual futures faces the risk that when they wish to unwind the trade, the spread may be arbitrarily large.

Our key insight is that a positive and even modestly large spread can still leave the arbitrageur with a positive payoff because funding payments also increase with the spread. As long as the spread does not diverge to infinity, there will be some future bounded unwinding time t when the accumulated funding rate payments overcome any finite potential losses at unwinding.

Note that if we relax the assumption that the short rate is constant for arbitrageurs, this may introduce some additional risk to the four parts of terms in Table 1: (1) mark-to-market changes in futures price; (2) funding-rate payment; (3) reinvestment in the spot market; (4) rollover of the short-term debt. This extension also relates to the literature examining the differences between futures and forwards in the context of stochastic interest rates, such as [Cox et al. \(1981\)](#) and [Grinblatt and Jegadeesh \(1996\)](#). They show that marking-to-market introduces an additional term in the futures price related to the covariance between the underlying asset and interest rate. Empirically, however, the volatility in the short rate is considerably smaller than the volatility in the deviation from futures to its fundamental price. Moreover, the changes in the short rate have opposite effects on payment from the cash market and the funding payment, as well as mark-to-market margin in the futures market and reinvestment in the spot market. Therefore, we abstract from short-rate randomness here but note it is an interesting avenue for future work.

For completeness, we next consider the edge case where for a given interest rate, the inverse of the funding rate function maps to an interval rather than a point. From Figure 3, we can see that this can happen on Binance only if the interest rate exactly equals ι .

Proposition 2 (No-arbitrage price: Edge Case). *Suppose $\Phi^{-1}(r)$ is an interval, under Assumptions 1, 2, and 3, there is an absence of random-maturity arbitrage opportunities if and only if*

$$F_t \in \left[\left(1 - \inf \{ \Phi^{-1}(r) \} \right)^{-1} S_t, \left(1 - \sup \{ \Phi^{-1}(r) \} \right)^{-1} S_t \right]. \quad (11)$$

Proof. The proof of this proposition is provided in Appendix E. □

Having derived no-arbitrage prices for perpetual futures for a general specification of the funding rate function Φ , we next specialize these results for the case of Binance, which is currently the largest perpetual futures exchange by volume and the source of our data below.

The funding rate on Binance can be written as:

$$\Phi\left(\frac{F-S}{F}\right) = \kappa \frac{F-S}{F} + \text{clamp}\left(\iota - \kappa \frac{F-S}{F}, -\gamma, \gamma\right), \quad (12)$$

where $\frac{F-S}{F}$ is the premium index of futures to spot, κ is a scaling constant for premium, $\text{clamp}(x, l, h) \equiv \min(\max(x, l), h)$ is the function that bound x between l and h , ι denotes the interest component in the funding rate, and γ is a parameter determining the width of the clamp. Currently on Binance, $\kappa = 1$, $\gamma = 0.05\%$, and $\iota = 0.01\%$. We plot the Φ as a function of the premium index $\frac{F-S}{F}$ in Figure 3.

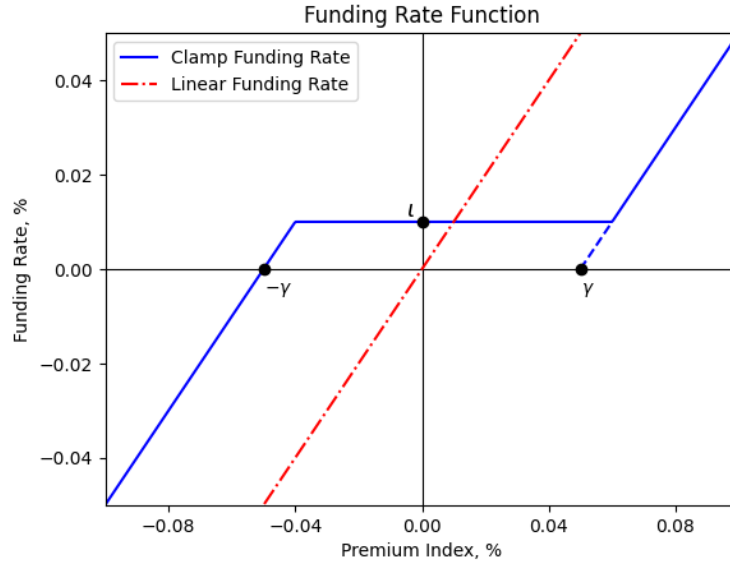


Figure 3 The Funding Rate Function Φ on Binance

This figure plots the funding rate as a function of the average premium index, $\frac{F-S}{F}$.

Corollary 2 (No-arbitrage price for Binance). *The funding rate function on Binance satisfies Assumption 3. Furthermore, under Assumptions 1 and 2, there is an absence of random-maturity arbitrage opportunities if and only if*

$$F_t = \frac{\kappa}{\kappa + \text{sgn}(\iota - r) \cdot \gamma - r} S_t, \quad \text{when } r \neq \iota, \quad (13)$$

$$F_t \in \left[\frac{\kappa}{\kappa + \gamma - r} S_t, \frac{\kappa}{\kappa - \gamma - r} S_t \right], \quad \text{when } r = \iota. \quad (14)$$

where $\text{sgn}(\cdot)$ is the sign function which equals $+1(-1)$ if its argument is positive (negative).

Proof. The proof follows from applying Proposition 1 and 2 to Binance funding rate function. We provide detailed scenario analysis in Appendix E, which also gives the divergence speed of the deviation in this specific case. $\square$

Empirically, the arbitrageurs face trading costs when opening or terminating their positions. Suppose they face a round-trip trading cost of C . We can derive the no random maturity arbitrage bounds for perpetual futures.

Proposition 3 (No-arbitrage bounds for Binance). *Under Assumption 1, 2, and 3, when there is a round-trip trading cost C of executing the trading strategy, the absence of random-maturity arbitrage opportunities implies that perpetual futures price will lie within the following bound relative to the spot:*

$$F_t \in \left[\frac{\kappa}{\kappa + \text{sgn}(\iota - r) \cdot \gamma - r} S_t - C, \frac{\kappa}{\kappa + \text{sgn}(\iota - r) \cdot \gamma - r} S_t + C \right], \quad \text{when } r \neq \iota, \quad (15)$$

$$F_t \in \left[\frac{\kappa}{\kappa + \gamma - r} S_t - C, \frac{\kappa}{\kappa - \gamma - r} S_t + C \right], \quad \text{when } r = \iota. \quad (16)$$

Proof. The proof follows the same logic as Corollary 2. The only difference is we need to subtract a cost of C from the total payoff to the arbitrageur. We provide detailed scenario analysis in Appendix E. $\square$

In the presence of trading costs, when the deviation of the perpetual futures price from λS_t , is larger than the round-trip trading costs C , arbitrageurs would have a strong incentive to trade perpetual futures toward the price λS_t . Proposition 2 also prescribes a trading strategy to exploit the futures-spot divergence in markets with different levels of trading costs. In the next section, we provide an empirical analysis of the futures-spot deviation and present arbitrage trading strategies motivated by our theory.

Before turning to the data, we pause to relate our results to two recent theoretical contributions. Angeris et al. (2022) consider an inverse problem to our setting. It assumes there is no arbitrage and that the perpetual futures price is a function of the spot, and derives the funding rate process that is consistent with the pricing formula. This takes a perspective of security design. However, empirically, funding rate mechanisms are predetermined and fixed for traders. Moreover, because of the limits of arbitrage, futures prices can deviate from the fixed function set by the designer. Our approach prescribes a strategy to exploit divergence from the fundamental price and offers a closer alignment with real-world trading conditions.

Compared to Akerer et al. (2023), our initial draft, disseminated December 2022 (see <https://arxiv.org/abs/2212.06888v1>), considers the case where the funding rate is a linear function of the premium index, as illustrated by the dashed line in Figure 3. It shows that the absence of random-maturity arbitrage opportunities implies the price of perpetual futures lies within the following bound relative to the spot:

$$|F_t - \left(1 + \frac{r}{\kappa}\right) S_t| \leq C, \quad (17)$$

and that in the case without trading costs, the price would be

$$F_t = \left(1 + \frac{r}{\kappa}\right) S_t. \quad (18)$$

Ackerer et al. (2023, <https://arxiv.org/abs/2310.11771v1>), dated October 2023, uses a different set of assumptions and solution method to derive no-arbitrage prices for a class of perpetual contracts. In their paper, the price relevant for the case we analyzed is:

$$F_t = \frac{\kappa}{\kappa - r} S_t. \quad (19)$$

Their paper further pointed out an inconsistency in the cash flow specification in our original derivation. After correcting this issue, Equation (19) above is consistent with their derivation in this case, and they deserve credit for deriving the correct formula first.³ However, their framework does not allow for the clamp feature, an empirically important component of the funding rate specifications employed by major perpetual futures exchanges.⁴

Another advantage of our derivation is that it makes explicit the arbitrage trading strategy in case of deviations and provides bounds in the case of trading costs which we use in our empirical analysis. The approach also enables the analysis of no-arbitrage trading for the clamped funding rate, a specification less understood by practitioners so far according to Han (2024).

Finally, we differ in our assumptions about price deviations. We assume that the *deviations* between the futures and spot are bounded, whereas Ackerer et al. (2023) assume that a transversality condition holds such that the futures price (and therefore the spot) grows at a slow enough rate relative to the funding payments intensity κ . But even under their assumptions, it can be shown that strategies aimed at arbitrage of deviations from their benchmark prices are necessarily random-maturity arbitrage strategies, as we emphasize here.

³We thank Damien Ackerer, Julien Hugonnier, and Urban Jermann for the comments that helped clarify the issue.

⁴According to Han (2024), the exchanges which have perpetual futures with the clamp feature account for more than 75% of open interest for Bitcoin perpetual futures.

4 Data and Empirical Analysis

We conduct an empirical analysis of perpetual futures arbitrage strategies. We first describe the data. We then measure the deviations of the crypto futures-spot spread from the no-arbitrage benchmark. Finally, we implement a trading strategy that exploits deviations from random-maturity no-arbitrage bounds and quantify the gains from this strategy.

4.1 Data

We focus on the five largest cryptocurrencies excluding stablecoins: Bitcoin (BTC), Ether (ETH), BNB (BNB), Dogecoin (DOGE), and Cardano (ADA) with a total market cap of \$1.83 trillion, which account for 73.2% of the total market share of the Crypto market by March 2024.

For each token, we obtain perpetual futures and spot prices at a 1-hour frequency from Binance. Binance is by far the leading exchange in the crypto realm. Another major benefit of using the Binance data is: nearly every part of our trading strategies can be completed within the same platform without delay in transferring funds. As such, it approximates the real-world investment opportunities faced by traders.

We retrieve the perpetual funding rates from Binance. The funding rate is paid every 8 hours on Binance. So we have futures and spot prices every hour and realized funding rate payment at 8:00, 16:00, and 0:00 GMT each day. The perpetual and spot tradings occur 24 hours per day and 7 days a week so there are no after-market hours in this market.

We get the earliest possible data on perpetual futures trading from Binance. The table below lists the starting and ending dates of our data for each cryptocurrency. Our data ends on 2024-03-11, covering the fallout of the FTX.

Asset	Start date	End date	N
BTC	2020-01-08	2024-03-11	36,578
ETH	2020-01-08	2024-03-11	36,578
BNB	2020-02-10	2024-03-11	35,777
DOGE	2020-07-10	2024-03-11	32,152
ADA	2020-01-31	2024-03-11	36,017

Table 2 Sample Descriptions

This table presents the sample start and end dates for the five cryptocurrencies and their total number of observations.

We obtain trading costs from Binance’s website⁵. In general, trading costs for the spot market are significantly larger than those for the perpetual futures for similar trading volume because futures are typically traded with leverage. Fee tiers are attributed to the 30-day trading volume. We attribute high trading costs with a 30-day spot trading volume above \$1 million and futures trading volume above \$15 million (small individual trader). Medium trading costs are attributed to a 30-day spot trading volume above \$150 million and futures trading volume above \$1 billion (small funds). Low trading costs are attributed to a 30-day spot trading volume above \$2 billion and futures trading volume above 12.5 billion (large funds). Zero fees can be negotiated with customized contracts, for example, for market makers. We consider trading costs for makers instead of takers because institutions typically trade maker orders.⁶

Table 3 presents the specification of different trading costs. It also shows the random-maturity arbitrage bound for the deviation between the perpetual futures and the no-arbitrage price (ρ_l and ρ_u). The bounds become wider as the trading costs increase. Detailed explanations of the trading costs specifications are also provided in Appendix Figure A.2 and Figure A.3.

⁵<https://www.binance.com/en/fee/trading> provides data on trading fees in perpetual futures and the spot market

⁶Takers trade market orders while makers trade limit orders. Takers take liquidity from the market while makers make the market or provide liquidity to the market.

Fee tier	Spot	Futures	ρ_l	ρ_u
No	0.0000%	0.0000%	−0.0%	0.0%
Low	0.0225%	0.0018%	−53.2%	53.2%
Medium	0.0450%	0.0072%	−114.3%	114.3%
High	0.0675%	0.0144%	−179.4%	179.4%

Table 3 Trading Costs Tiers

Fee tiers are assigned based on past 30-day trading volume. High fees correspond to a 30-day trading volume above \$1mn in spot and above \$15mn in perpetuals, typically an individual trader. Medium fees attribute to a 30-day trading volume above \$150mn in spot and above \$1bn in perpetuals (small funds). Low fees are attributed to a 30-day trading volume above \$2bn in spot and above \$12.5bn in perpetuals (large funds). The no-fee tier can be negotiated with customized contracts, for example for market makers. We also report the theory implied no arbitrage upper and lower bounds (ρ_l and ρ_u) for ρ under different trading cost specifications, calculated using Equation 23.

In addition to the direct trading fees, traders also face bid-ask spread and price impact costs in our arbitrage strategies. To assess the importance of such indirect trading costs, we attain from Kaiko a high-frequency sample with one month of order book snapshots taken every 30 seconds. With this data, we estimate that the average bid-ask spread for perpetual futures is less than 0.05 bps, and the price impact of a trade with a size equal to 0.5 million is less than 0.3 bps. Combined, these indirect costs are less than 0.35 bps, which is substantially smaller than our direct trading fees. Therefore, we focus our analysis on the direct trading fees, as they are the dominant form of trading costs for our arbitrage strategies.

To measure the deviation from the no-arbitrage price, we also obtain the interest rate data from Aave, a leading open-source DeFi liquidity protocol. Customers on Aave can either be suppliers or borrowers of cryptocurrencies. Because of the anonymity and decentralization of the DeFi system, all borrowings are over-collateralized and the collateral can serve as an additional supply to the system for borrowing. The customers can also supply spare currencies to the system to earn an interest rate. The supply interest rate is typically different from the borrowing interest rate. Both interest rates and

their wedge are determined algorithmically based on the market condition of supply and demand. We use interest rates from this platform because we believe it is a good proxy for the funding condition in the crypto market. Appendix D shows that our results are robust to using the interest rate from the traditional financial market or using the margin borrowing rate from Binance.

Our theory indicates using a risk-free rate available to the arbitrageurs. Therefore, we consider the interest rates on the stablecoins, which are not subject to volatile spot price movement and are the currency of denomination for perpetual futures margins. There are three major stablecoins traded on Aave: USDT, USDC, and DAI. To get a robust measure of the risk-free rate, we take an average of the three interest rates to arrive at our final risk-free supply and borrowing rate for the arbitrageurs.

We plot the time-series evolution of the interest rate in the Appendix Figure A.4. During the early years, we see higher interest rate volatility due to the funding liquidity of the DeFi platform. In later samples, interest rates are more stable and approach interest rates in traditional financial markets.

Our interest rate data starts from 2020-01-08. Therefore, for coins with perpetual data available before the time (BTC and ETH), we begin the analysis from 2020-01-08. For other coins (BNB, DOGE, ADA), we begin the analysis from the time when they have data available.

4.2 Deviations of perpetual futures from no-arbitrage benchmarks

The focus of our empirical analysis is the annualized deviation ρ , defined as the interest rate spread that would rationalize an observed future-spot spread. Under the clamp specification of funding rate from Binance, we have:

$$F = \frac{\kappa}{\kappa + \text{sgn}(t - r) \cdot \gamma - (r + \rho)} S. \quad (20)$$

For example, an estimated ρ of one percentage point would mean that borrowing costs in the cash market faced by arbitrageurs would have to be one percentage point higher than the prevailing interest rate r in the cash market for the futures-spot gap to be arbitrage-free.

Rearranging Equation (20), we obtain the following equation for ρ :

$$\rho = \kappa \left(\frac{F - S}{F} \right) + \text{sgn}(\iota - r) \gamma - r. \quad (21)$$

This equation shows that deviations are closely related to the premium index $\frac{F-S}{F}$, adjusted for the clamp parameter γ and the interest rate r . This definition is the same in spirit to [Du et al. \(2018\)](#), who measure covered interest parity (CIP) deviations with the wedge that would equate the dollar borrowing rate and the synthetic dollar borrowing rate.⁷

Furthermore, for each tier of trading cost in Table 3, we need to calculate the corresponding threshold value ρ . We map ρ to a trading cost using the bound from Proposition 2, specialized for a proportional trading cost c ,

$$F - \frac{\kappa}{\kappa + \text{sgn}(\iota - r) \gamma - r} S = cF. \quad (22)$$

Equations 20 and 22 together imply we can express the threshold ρ as:

$$\rho = (\kappa + \text{sgn}(\iota - r) \gamma - r) c \approx \kappa c, \quad (23)$$

where the last approximation follows from the empirical fact that κ is much larger than

⁷Alternatively, we may define ρ as the wedge that satisfies

$$F = \frac{\kappa}{\kappa + \text{sgn}(\iota - (r + \rho)) \cdot \gamma - (r + \rho)} S.$$

Using this definition, after rearranging terms, ρ simplifies to:

$$\rho = \kappa \left(\frac{F - S}{F} \right) + \text{clamp}(\iota - \kappa \frac{F - S}{F}, -\gamma, \gamma) - r = \Phi - r.$$

This result is closely related to the key equation used to determine the fundamental price, $\Phi = r$. We adopt the first definition for its greater tractability when incorporating trading costs, but both have their merits.

γ and r .⁸ Therefore, we calculate the upper and lower trading threshold in Table 3 as κc and $-\kappa c$.

Each hour, we calculate ρ defined in Equation (21) using data on the perpetual futures price, spot price, and the crypto risk-free interest rate. When the perpetual futures price is above the spot, an arbitrageur would short the futures and long the spot, and would finance her position by borrowing in the cash market (as is shown in Table 1). On the other hand, when the perpetual futures price is below the spot, an arbitrageur would long the futures, short the spot, and invest the proceeds from shorting in the cash market (as is shown in Table 1).

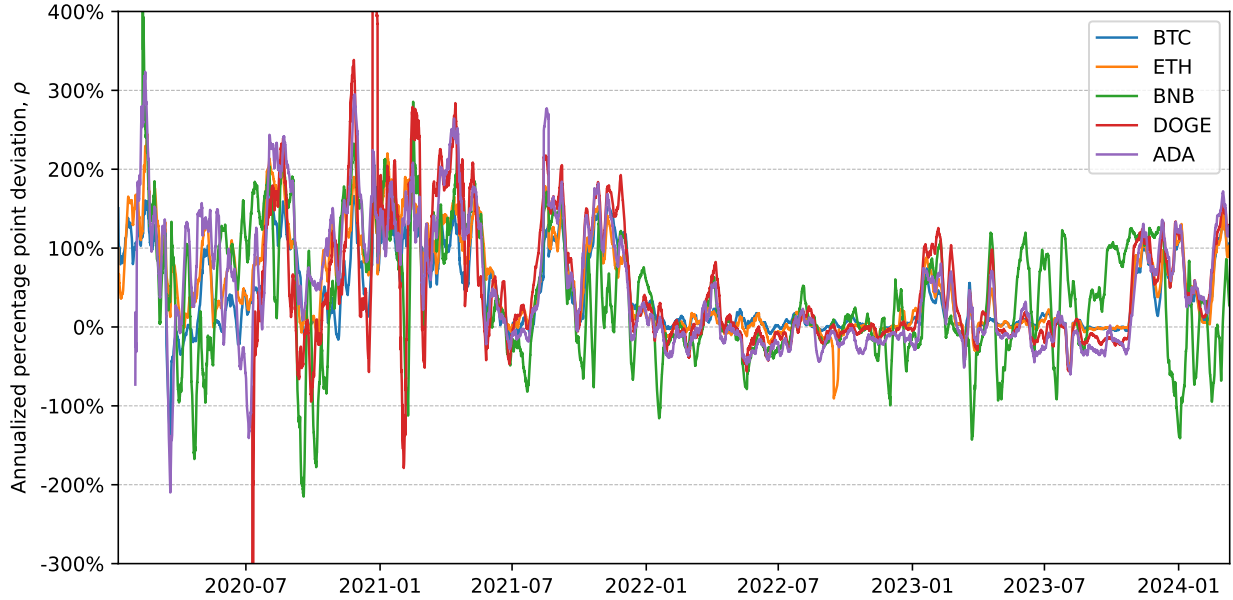


Figure 4 Deviations of Perpetual Futures from No-arbitrage Benchmarks

This figure presents the 7-day moving averages of the annualized deviation of the perpetual futures-spot spread from the no-arbitrage benchmark for the five cryptocurrencies.

Table 4 reports summary statistics for ρ in our data. As can be seen, the deviation from the no-arbitrage price is predominantly positive across a substantial portion of the sample, suggesting perpetual futures are more expensive than the no-arbitrage price on average.

⁸Incorporating γ and r makes a negligible difference for the threshold ρ .

However, there is considerable variation in these deviations, with the standard deviation exceeding twice the mean deviation for most tokens. The mean absolute deviation is about 60% to 90% per year across different cryptocurrencies.

Panel A: Full Sample								
Asset	ρ				$ \rho $			
	Mean	Median	Std	p-value	Mean	Median	Std	p-value
BTC	0.42	0.19	0.84	0.00	0.52	0.25	0.78	0.00
ETH	0.52	0.26	0.88	0.00	0.63	0.33	0.81	0.00
BNB	0.38	0.31	1.23	0.00	0.90	0.68	0.92	0.00
DOGE	0.50	0.23	1.90	0.00	0.86	0.49	1.77	0.00
ADA	0.51	0.27	1.27	0.00	0.84	0.53	1.08	0.00

Panel B: Subsample Comparison								
Asset	ρ Before 2022				ρ After 2022			
	Mean	Median	Std	p-value	Mean	Median	Std	p-value
BTC	0.69	0.62	1.08	0.00	0.17	0.08	0.42	0.00
ETH	0.92	0.85	1.01	0.00	0.16	0.06	0.53	0.00
BNB	0.69	0.81	1.53	0.00	0.11	0.07	0.81	0.09
DOGE	0.99	0.95	2.83	0.00	0.18	0.06	0.62	0.01
ADA	0.99	0.94	1.60	0.00	0.09	0.03	0.66	0.11

Table 4 Summary Statistics for Deviations from No-arbitrage Prices

The table summarizes statistics for ρ across various crypto tokens. Panel A presents the summary statistics of ρ and $|\rho|$ across the whole sample. Panel B presents the summary statistics for ρ before and after 2022-01-01. p-value corresponds to the Newey-West adjusted p-value, testing the null hypothesis that the corresponding mean of ρ is zero. The sample periods are shown in Table 2.

Subsample analysis reveals that the mean deviation and variation in deviation shrink over time. After 2022, the average deviation reduces by almost 80% with volatility shrinking by more than 50% for most tokens. Figure 4 depicts the 7-day moving average of ρ for each of the five examined cryptocurrencies. It reveals considerable comovement in futures-spot spreads across these cryptocurrencies, suggesting a common factor at play that drives the observed divergence. This observation is corroborated by Table 5, which demonstrates high correlations among futures-spot deviations across the different cryp-

tocurrencies. This trend underscores the interconnectedness of cryptocurrency markets and is consistent with the comovement of traditional currencies documented by [Lustig, Roussanov, and Verdelhan \(2011\)](#).

One hypothesis for this significant comovement in ρ across various cryptocurrencies pertains to time-varying funding constraints encountered by arbitrageurs in the market ([Brunnermeier and Pedersen, 2009](#); [Garleanu and Pedersen, 2011](#)). Because arbitrageurs often serve as marginal traders in all markets, their funding constraints are potentially reflected in the ρ across all major cryptocurrency markets. Our theoretical and empirical analyses underscore the importance of measuring ρ . Doing so not only sheds light on potential market stress but also provides a gauge for the shadow costs associated with the trading and funding constraints arbitrageurs face.

Conversely, from the demand side, shared sentiment across various cryptocurrencies could also contribute to explaining the concurrent movement of ρ across different markets. Our theoretical analysis suggests that arbitrageurs will only meet the market demand if the price deviation surpasses the combined trading and funding costs. As a result, the overall market sentiment in the futures market relative to the spot market becomes evident in the price difference between futures and spot prices. If the sentiment across different markets is influenced by a common factor, we would anticipate a significant comovement in the futures-spot spread.

Moreover, arbitrageurs are willing to absorb the idiosyncratic demand first because combining the arbitrage trades across different idiosyncratic deviations would generate a high Sharpe ratio strategy as in [Kozak et al. \(2018\)](#). Therefore, in equilibrium, after arbitrageurs have absorbed the idiosyncratic demand, we anticipate the emergence of a common factor in deviations across different cryptocurrencies, driven by limits to arbitrage.

In Section 5, we delve deeper into explaining the futures-spot spread. Our results

		Arbitrage Deviation				Funding Rate					Spot Return				
		ρ_{ETH}	ρ_{BNB}	ρ_{DOGE}	ρ_{ADA}	fr_{BTC}	fr_{ETH}	fr_{BNB}	fr_{DOGE}	fr_{ADA}	R_{BTC}	R_{ETH}	R_{BNB}	R_{DOGE}	R_{ADA}
Arbitrage Deviation	ρ_{BTC}	0.88	0.53	0.73	0.79	0.67	0.58	0.45	0.40	0.53	-0.02	0.01	0.03	0.09	0.02
	ρ_{ETH}		0.58	0.69	0.82	0.64	0.72	0.48	0.41	0.58	0.06	0.08	0.07	0.11	0.08
	ρ_{BNB}			0.50	0.53	0.45	0.45	0.82	0.37	0.43	0.01	0.05	0.03	0.05	0.06
	ρ_{DOGE}				0.70	0.46	0.46	0.41	0.61	0.50	0.07	0.09	0.11	-0.00	0.07
	ρ_{ADA}					0.57	0.56	0.44	0.45	0.73	0.07	0.10	0.10	0.10	0.08
Funding Rate	fr_{BTC}						0.86	0.65	0.63	0.74	0.08	0.09	0.15	0.13	0.10
	fr_{ETH}							0.63	0.58	0.74	0.12	0.16	0.18	0.17	0.14
	fr_{BNB}								0.55	0.60	0.04	0.09	0.11	0.10	0.10
	fr_{DOGE}									0.68	0.12	0.14	0.23	-0.01	0.13
	fr_{ADA}										0.10	0.15	0.21	0.16	0.12
Spot Return	R_{BTC}											0.84	0.71	0.46	0.70
	R_{ETH}												0.73	0.43	0.74
	R_{BNB}													0.34	0.65
	R_{DOGE}														0.42

Table 5 Correlation of arbitrage deviation (ρ), funding rate (fr), and spot returns (R)

This table presents the correlation among different cryptocurrencies' futures-spot deviation ρ (rows 3 to 7), funding rate (rows 8 to 12), and spot returns (rows 13 to 16).

demonstrate that the past returns of each cryptocurrency serve as significant explanatory variables for the time-series variation in the spread.

Interestingly, the correlation between deviations and spot market returns is quite modest. Although spot market returns are highly correlated among themselves, in line with the strong market factor results from Liu et al. (2022), and no-arbitrage price deviations are highly correlated among themselves, it appears that different forces drive these distinct phenomena.

Furthermore, we find that after the year 2022, futures-spot spreads become much closer and anchored to 0 compared to earlier samples. This suggests the market is becoming increasingly efficient.

Last but not least, by design, ρ is also highly correlated with the funding rate. The funding rate does not correlate perfectly with ρ because in real-world implementations: (1) the funding rate is calculated as a weighted average of futures-spot price deviations, with larger weight given to more recent observations; (2) in calculating the funding rate, Binance does not just consider the quote price, they also use an impact margin notional to consider price impact of the trade.⁹

4.3 Random-maturity Arbitrage Strategy

We next provide a trading strategy motivated by our random-maturity arbitrage theory. Table 3 reports for different trading cost tiers, the bounds (ρ_l and ρ_u) beyond which there exist random-maturity arbitrage opportunities. We consider a simple trading strategy: whenever ρ enters the region outside the annualized round-trip trading costs in Table 3, we open the position. We close the position when ρ first goes back to 0.

Figure 5 provides an illustration of the trading strategy. When the deviation is above

⁹See <https://www.binance.com/en/support/faq/360033525031> for details of the funding rate calculation.

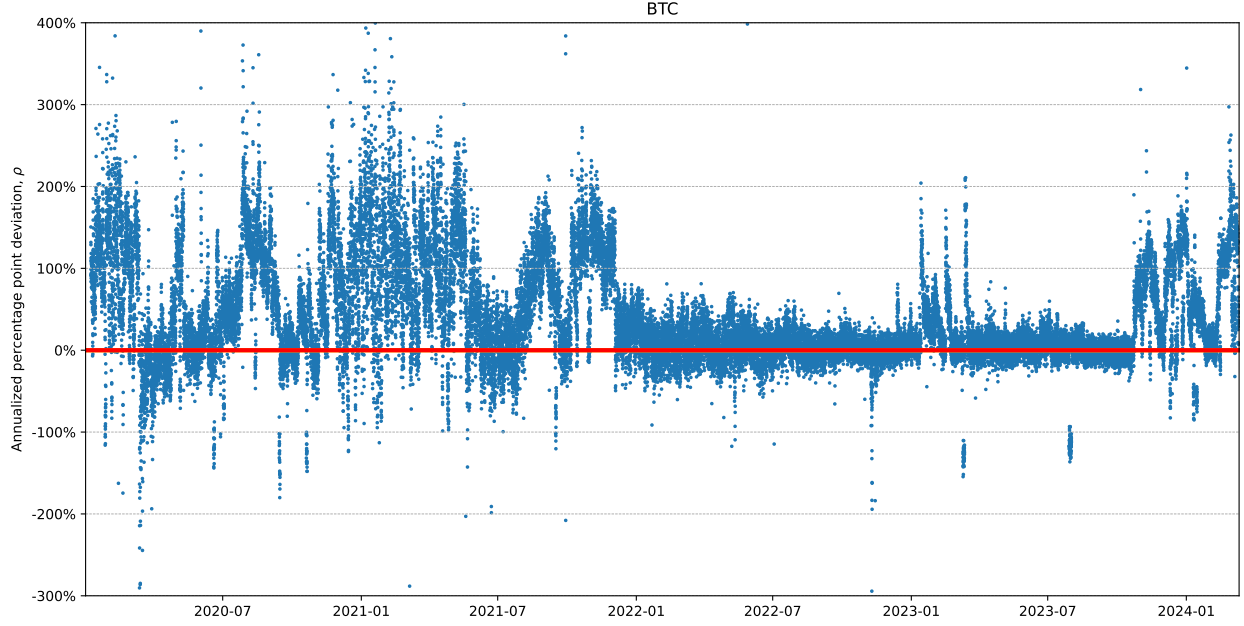


Figure 5 Random-maturity Arbitrage Strategy for Bitcoin

Deviation from no-arbitrage price (ρ) and trading rules of the random-maturity arbitrage strategy we implement for BTC. Each blue dot in the figure represents the deviation of futures from the no-arbitrage price calculated using Equation 21 at hourly frequency. The number is annualized for ease of interpretation. The red line represents the zero benchmark.

the red line, the strategy opens a long position in the spot and a short position in perpetual. Conversely, when the deviation is below the red line, the strategy is short the spot and long the perpetual.

Table 6 presents the performance of the trading strategy over the years. We find that the trading strategy generates a high Sharpe ratio for all five tokens. The large returns with low volatility and drawdowns suggest that our theory provides useful benchmarks. A deviation from the no-arbitrage price indeed implies a near-arbitrage opportunity.

In Table 7, we compare the performance of our baseline strategy with three alternative trading strategies. The first alternative ignores the clamp feature and relies on deviations from the price (18) provided in our earlier draft, or equivalently, in Akerer et al. (2023). We find that our proposed strategy almost doubles the Sharpe ratio for Bitcoin arbitrage and consistently improves the arbitrage performance for the other currencies. This

		2020	2021	2022	2023	2024	All
BTC	SR	7.18	12.36	37.85	15.37	11.84	11.65
	Return	49.62	64.81	60.02	41.68	21.44	52.55
	Volatility	6.91	5.24	1.59	2.71	1.81	4.51
	MaxDD	-3.82	-2.00	-0.36	-0.26	-0.17	-3.82
	OtC time	8.35	9.17	2.97	3.71	11.44	4.92
	N	8,616	8,760	8,760	8,760	1,682	36,578
ETH	SR	10.43	12.04	25.92	12.82	13.90	12.77
	Return	67.53	75.56	70.73	44.81	25.11	62.83
	Volatility	6.48	6.27	2.73	3.49	1.81	4.92
	MaxDD	-2.62	-2.28	-0.67	-0.29	-0.16	-2.62
	OtC time	11.19	8.80	3.02	3.68	8.54	5.05
	N	8,616	8,760	8,760	8,760	1,682	36,578
BNB	SR	15.30	16.89	34.91	17.66	22.64	17.66
	Return	159.04	166.86	134.73	87.08	102.83	134.74
	Volatility	10.40	9.88	3.86	4.93	4.54	7.63
	MaxDD	-1.96	-1.66	-0.83	-0.97	-0.25	-1.96
	OtC time	8.50	6.44	3.32	6.33	6.60	5.46
	N	7,814	8,760	8,760	8,760	1,682	35,776
DOGE	SR	18.14	12.78	45.37	19.58	12.29	14.90
	Return	409.29	207.91	191.43	62.89	35.36	181.12
	Volatility	22.56	16.27	4.22	3.21	2.88	12.16
	MaxDD	-13.90	-5.61	-0.29	-1.65	-0.29	-13.90
	OtC time	3.90	6.91	2.57	4.60	12.19	4.13
	N	4,189	8,760	8,760	8,760	1,682	32,151
ADA	SR	14.11	13.07	45.97	34.87	13.03	19.76
	Return	147.63	138.28	210.28	145.53	42.29	155.16
	Volatility	10.46	10.58	4.57	4.17	3.25	7.85
	MaxDD	-3.47	-5.69	-0.24	-0.50	-0.18	-5.69
	OtC time	7.74	6.28	2.70	3.46	13.90	4.32
	N	8,054	8,760	8,760	8,760	1,682	36,016

Table 6 Performance of Random Maturity Arbitrage Trading Strategy Over Time

This table presents the Sharpe ratios, annualized returns (%), standard deviations (%), maximum drawdowns (%), and the average open-to-close time (hours) of the unrestricted random-maturity arbitrage trading strategies for five different cryptocurrencies with high trading costs. The returns reported are excess returns from the random maturity arbitrage strategies.

suggests that taking the clamp feature into account is economically important in practice.

In the early years of our sample, the infrastructure for shorting cryptocurrencies was underdeveloped. To address this concern, we consider a long-spot-only strategy, in which we set the portfolio weights at 0 if the strategy shorts the spot and longs the futures. We find that even with this restriction, the strategy still generates a high Sharpe ratio in all cryptocurrencies.

The last alternative we consider is based on [Christin et al. \(2022\)](#) which proposes a carry strategy in the perpetual futures market where the trader longs the spot, shorts the perpetual and holds on to the strategy throughout the sample. We present the performance of this strategy in the last column of Table 7. We find that the long-only strategy significantly outperforms the carry strategy. These results highlight the usefulness of our fundamental price. Given this benchmark, we can identify deviation in real-time and dynamically rebalance the position to take advantage of the deviation rather than making a directional bet as in the Carry strategy.

In Appendix C, we also present the performance of a data-driven strategy that optimally determines the open and close thresholds using data from the past 6 months. The data-driven strategy has a performance similar to or worse than our theory-motivated strategy, suggesting that our proposed trading strategy is close to being optimal for arbitrage.

In the presence of trading costs, as is motivated by our theory, we consider the threshold trading rules presented in Table 3. We visualize the trading strategies in Figure A.5 in Appendix A. Different trading strategies have the same close-position line, which is equal to 0, while the open-position line adjusts to the level of trading costs.

Since our trading strategy under trading costs is a threshold trading rule, to annualize Sharpe ratios we follow [Lucca and Moench \(2015\)](#), which considers a trading strategy that is only active on FOMC announcement dates. We first calculate the mean (μ) and standard deviation (σ) of our trading strategy during the time it is active. Next, we scale μ/σ by the number of periods the strategy is active in a year:

$$SR = \frac{\mu}{\sigma} \sqrt{N_a},$$

where μ and σ are average hourly returns and N_a is the average number of hours the strategy is active in a year. We follow the same approach to annualize the returns and

		Trading Strategies			
		Correct	No Clamp	Long-only	Carry
BTC	SR	11.65	6.06	8.20	2.24
	Return	52.55	27.52	31.37	10.18
	Volatility	4.51	4.54	3.83	4.55
	MaxDD	-3.82	-3.82	-3.82	-5.33
	α	58.82	31.22	33.30	7.89
	t_α	19.69	11.09	18.39	4.96
	Active %	100.00	100.00	74.05	100.00
	OtC time	4.92	16.23	7.28	-
ETH	SR	12.77	8.29	8.64	2.66
	Return	62.83	40.98	38.02	13.22
	Volatility	4.92	4.94	4.40	4.96
	MaxDD	-2.62	-2.62	-2.62	-2.80
	α	72.39	49.86	41.45	10.77
	t_α	17.45	11.02	16.21	5.31
	Active %	100.00	100.00	74.61	100.00
	OtC time	5.05	13.60	7.53	-
BNB	SR	17.66	14.71	11.28	-0.66
	Return	134.74	112.80	64.80	-5.14
	Volatility	7.63	7.67	5.74	7.76
	MaxDD	-1.96	-1.96	-1.66	-33.61
	α	145.62	116.29	68.54	-7.13
	t_α	13.94	11.91	12.54	-2.21
	Active %	100.00	100.00	63.30	100.00
	OtC time	5.46	7.46	6.91	-
DOGE	SR	14.90	12.68	9.29	1.01
	Return	181.12	154.69	96.76	12.39
	Volatility	12.16	12.20	10.41	12.31
	MaxDD	-13.90	-13.90	-13.90	-14.37
	α	221.37	195.57	112.52	6.94
	t_α	11.27	9.13	11.91	2.80
	Active %	100.00	100.00	64.91	100.00
	OtC time	4.13	6.81	5.36	-
ADA	SR	19.76	15.78	12.13	1.64
	Return	155.16	124.91	84.16	13.15
	Volatility	7.85	7.92	6.94	8.03
	MaxDD	-5.69	-5.69	-5.69	-6.58
	α	176.12	144.73	91.75	9.17
	t_α	20.12	16.38	20.67	3.92
	Active %	100.00	100.00	65.10	100.00
	OtC time	4.32	6.21	5.63	-

Table 7 Comparison of Different Arbitrage Trading Strategies

This table compares the performance of 4 trading strategies. Correct is the strategy proposed in this paper. No Clamp is the strategy with the fundamental price given in our earlier draft and [Akerer et al. \(2023\)](#) which ignores the clamp feature. Long-only is our strategy with a restriction of no short sale in the spot market. Carry is the strategy studied in [Christin et al. \(2022\)](#). Statistics reported are the annualized Sharpe ratio, return (%), volatility (%), max drawdown (%), alpha (%), t-stat of the alpha, proportion of time the strategy is active (%), and average open-to-close (OtC) position duration in hours.

standard deviations: $\mu_{ann} = \mu N_a$, $\sigma_{ann} = \sigma \sqrt{N_a}$.

Table 8 presents the strategy’s performance under different fee tiers. As trading costs increase, the random-maturity arbitrage strategy engages in less active trading. As a result, the average duration of open-to-close time increases. However, even for the high trading cost tier, we find significant and large Sharpe ratios across different tokens. The strategy also delivers highly significant alphas relative to the 3-factor model by [Liu et al. \(2022\)](#) and the 5-factor model by [Cong et al. \(2022a\)](#). Evidently, the strategy’s performance cannot be explained by previously suggested risk factors.

Comparing the results with [Du, Tepper, and Verdelhan \(2018\)](#), we find that deviations in crypto perpetual futures are considerably larger in magnitude. As a result, gains from the arbitrage strategies we study are also larger. Even though the volatility of the trading strategy also scales up, the Sharpe ratios in the crypto space are still larger than those in the traditional foreign exchange market, as reported in [Du, Tepper, and Verdelhan \(2018\)](#).

When a futures-spot gap opens up, gains from the trading strategy could potentially arise from two main sources: price convergence and funding rate payments. While industry publications usually emphasize the funding rate channel, we note that price convergence can generate quicker gains from arbitrage if dislocations are short-lived. To examine these two sources empirically, in Table 9, we provide a decomposition of the trading strategy’s performance into price convergence versus funding rate payment. We find that price convergence plays a dominant role in total trading returns, while funding rate payments have a more minor role, which seems to diminish over time.

The success of the trading strategy supports the theory of random-maturity arbitrage. Whenever there is a deviation larger than the gap implied by the theory, betting on convergence tends to generate a positive payoff at some uncertain future time. Therefore, the convergence arbitrage trade generates high Sharpe ratios.

		Fee Tiers			
		No	Low	Medium	High
BTC	SR	11.65	4.30	3.38	3.35
	Return	52.55	18.04	13.88	12.96
	Volatility	4.51	4.19	4.10	3.87
	MaxDD	-3.82	-3.82	-3.89	-2.63
	α	58.82	16.98	12.62	12.04
	t_α	19.69	7.39	6.02	5.83
	Active %	100.00	45.44	33.69	24.19
	OtC time	4.92	40.74	89.94	96.16
ETH	SR	12.77	5.83	4.69	4.56
	Return	62.83	26.38	20.55	18.81
	Volatility	4.92	4.52	4.38	4.12
	MaxDD	-2.62	-2.67	-2.66	-2.64
	α	72.39	26.96	20.13	18.20
	t_α	17.45	7.41	6.35	5.62
	Active %	100.00	51.60	39.83	26.93
	OtC time	5.05	33.89	73.21	80.74
BNB	SR	17.66	11.41	8.35	6.93
	Return	134.74	81.83	56.43	43.68
	Volatility	7.63	7.18	6.75	6.31
	MaxDD	-1.96	-1.97	-2.02	-2.08
	α	145.62	72.51	46.99	34.95
	t_α	13.94	8.34	6.28	5.74
	Active %	100.00	73.41	52.28	38.55
	OtC time	5.46	10.72	21.40	29.03
DOGE	SR	14.90	9.22	6.27	4.97
	Return	181.12	108.46	71.35	55.22
	Volatility	12.16	11.77	11.38	11.10
	MaxDD	-13.90	-13.90	-13.90	-13.90
	α	221.37	116.91	66.46	44.87
	t_α	11.27	7.16	5.41	4.72
	Active %	100.00	60.85	40.48	30.13
	OtC time	4.13	7.62	12.68	16.28
ADA	SR	19.76	10.72	6.03	4.14
	Return	155.16	78.51	41.77	27.42
	Volatility	7.85	7.32	6.92	6.62
	MaxDD	-5.69	-5.69	-5.69	-5.69
	α	176.12	77.94	36.32	22.88
	t_α	20.12	13.57	8.79	6.23
	Active %	100.00	66.42	42.81	30.85
	OtC time	4.32	8.47	19.39	42.24

Table 8 Performance of Random Maturity Arbitrage across Different Trading Costs

Performance under different trading cost tiers. The fees for spot (futures) are 2.25 (0.18) bps, 4.5 (0.72) bps, and 6.75 (1.44) bps for the low, medium, and high trading cost levels. Statistics reported are the annualized Sharpe ratio, return (%), volatility (%), max drawdown (%), alpha (%), t-stat of the alpha, proportion of time the strategy is active (%), and average open-to-close (OtC) position duration in hours. The 3-factor data from [Liu et al. \(2022\)](#) used to calculate alpha ends on April 9th, 2023.

		2020	2021	2022	2023	2024	All
BTC	Return	49.62	64.81	60.02	41.68	21.44	52.55
	Price	29.99	34.76	55.93	36.22	3.30	37.61
	Funding	19.63	30.05	4.09	5.46	18.15	14.94
ETH	Return	67.53	75.56	70.73	44.81	25.11	62.83
	Price	41.98	40.66	65.52	39.17	7.54	45.05
	Funding	25.55	34.91	5.21	5.64	17.57	17.78
BNB	Return	159.83	166.86	134.73	87.08	102.83	134.91
	Price	125.07	132.19	124.80	73.48	74.90	111.75
	Funding	34.77	34.67	9.93	13.61	27.93	23.16
DOGE	Return	408.14	207.91	191.43	62.89	35.36	180.96
	Price	393.74	165.86	188.89	57.44	12.59	164.26
	Funding	14.39	42.05	2.54	5.45	22.78	16.70
ADA	Return	147.47	138.28	210.28	145.53	42.29	155.13
	Price	117.41	98.57	206.19	140.72	19.16	135.50
	Funding	30.06	39.71	4.09	4.80	23.13	19.62

Table 9 Return Decomposition: Price Convergence versus Funding Rate Payment

This table decomposes the portfolio return into the part due to price convergence and the part due to funding rate payment. The returns reported are excess returns from the random-maturity arbitrage strategies.

4.4 Liquidity Risk

Our theoretical and empirical analysis demonstrates substantial deviation from no-arbitrage prices of perpetual futures. These deviations could potentially be explained by liquidation risk faced by arbitrageurs. In this section, we first explore liquidation risk from a theoretical standpoint. We then present empirical evidence showing that, in practice, such liquidation risk is minimal.

Our analysis is related to [Alexander, Deng, and Zou \(2023\)](#) which analyzes the role of leverage and margin selection in mitigating forced liquidation for Bitcoin futures. Unlike their setting, we focus on the liquidation risk facing the arbitrageurs conducting random maturity arbitrage, which aims to bring futures prices back to their fundamental levels.

Suppose the arbitrageur has arbitrage capital M to engage in random maturity arbi-

trage, and her margin accounts in the spot and futures markets are linked.¹⁰ To avoid early liquidation, the real-time loss to the arbitrageur should never exceed M . For now, we focus on the case when $\iota < r$ and note that a similar argument would apply to the case when $\iota > r$.

Scenario 1: When $F_0 > \lambda S_0$, the deviation lies in the upper region of the Φ function. The no early liquidation condition states that the payoff defined in Equation (7) should be larger than $-M$:

$$-e^{-rt}u_t + u_0 + (\kappa - \gamma - r) \int_0^t u_s e^{-rs} ds \geq -M. \quad (24)$$

Let $\bar{u}_t$ denote the upper bound process implicitly defined by $e^{-rt}\bar{u}_t = u_0 + M + (\kappa - \gamma - r) \int_0^t \bar{u}_s e^{-rs} ds$, which provides an upper bound for all processes satisfying Inequality (24).

We can solve for $\bar{u}_t$ as:

$$\bar{u}_t = e^{(\kappa - \gamma)t}(u_0 + M),$$

which grows exponentially. As time increases, because of the cumulative funding rate payment, the upper bound becomes less and less binding.

We may ignore the funding accumulation and focus on the initial deviation. When conducting long-short arbitrage, as long as the futures-spot deviation does not have large changes, traders can avoid liquidation risk even with modest capital.

In this scenario, if $u_t \leq u_0 + M$, then the arbitrageur will never face early liquidation. Intuitively, the greater the initial deviation u_0 and the margin posted M , the lower the risk of liquidation.

Scenario 2: When $F_0 < \lambda S_0$, consider the case when the deviation falls into the clamp region of Φ function. Similarly, the payoff defined in Equation (9) should be larger than

¹⁰ Binance allows traders to transfer funds between their Spot and Futures wallet instantaneously and without any fees.

$-M$:

$$e^{-rt}u_t - u_0 - \int_0^t e^{-rs}(\mathfrak{t} - r)F_s ds \geq -M, \quad (25)$$

which implies a lower bound $\underline{u}_t$ for all processes satisfying the inequality:

$$\underline{u}_t = u_0 - M + \int_0^t e^{-rs}(\mathfrak{t} - r)F_s ds.$$

Similar to *Scenario 1*, we may ignore the funding cumulation and note that as long as $u_t \geq u_0 - M$, the arbitrageurs can avoid early liquidation.

The analysis highlights the changes in deviation as an important source of risk for early liquidation. Empirically, we find that the change is small most of the time and that the risk of early liquidation can be avoided with a modest amount of capital. We may see this through the maximum drawdown statistics reported in Table 6 and Table 8.

For BTC, for example, the maximum drawdown is above -4% across different specifications. It means that during our sample period, an arbitrageur could completely avoid early liquidation with only 4% of capital. Similar findings hold for other cryptocurrencies.

Our results demonstrate that the no liquidation risk constraint imposes a bound on the deviation from fundamental prices. Empirically, this bound is mild, allowing liquidation risk to be entirely avoided with a modest amount of capital in random maturity arbitrage.

5 Explaining Futures-spot Deviations

From Figure 4, we observe a strong common comovement of the futures-spot deviation across all crypto assets. Our goal is to understand the fundamental forces driving this common factor. We consider two potential hypotheses: (1) the time-varying funding constraints of arbitrageurs and (2) the time-varying relative demand from end-users for perpetual futures compared to the spot.

Both forces could potentially explain these patterns. Arbitrageurs will accommodate the relative demand from end-users only until the price deviation lies within the random-maturity no-arbitrage bound, which depends on arbitrageurs' funding conditions. As arbitrageurs are likely the marginal investors in all cryptocurrency markets, their time-varying funding constraints may create common time-series variation in ρ across different cryptocurrencies.

On the other hand, when the deviation lies within the no-arbitrage bound, variations in demand from the perpetual futures market compared to the spot market will influence the price deviation. It is plausible that relative demand has common factors across different cryptocurrencies. For instance, sentiment could drive their common variation, as perpetual futures allow for high leverage, which attracts overconfident, extrapolative, and sentiment-driven investors.

Determining which of the two factors better explains the observed price deviation is an empirical question. To shed light on this, we use past returns as a proxy for relative extrapolative demand in the perpetual futures market compared to the spot market. Past returns correlate with the demand from investors with extrapolative beliefs, who are more likely to trade in the perpetual futures market, given the high leverage it provides. Therefore, past returns would correlate with the relative demand in the perpetual futures market compared to the spot.

For the time-varying funding constraint, we consider using crypto return volatility as a proxy because, as in [Adrian and Shin \(2010\)](#), arbitrageurs likely face Value-at-Risk (VaR) type constraints, which are more likely to bind when market volatility is high.

Moreover, from Figure 4, we see a downward trend for the magnitude of the deviation. [Kondor \(2009\)](#)'s model of convergence trading predicts that greater competition among arbitrageurs reduces the profitability of arbitrage opportunities and deviations approach a martingale. Therefore, we add a time trend variable to the regression to test if the market

becomes more efficient as more arbitrageurs and capital compete in this market.

In sum, the regression models can be represented as:

$$\rho_{i,t} = \beta_{0,i} + \beta_1(t/365) + \beta_2 R_{i,t-120:t-1} + \beta_3 \sigma_{i,t-120:t-1} + \varepsilon_{i,t}, \quad (26)$$

$$|\rho_{i,t}| = \beta_{0,i} + \beta_1(t/365) + \beta_2 R_{i,t-120:t-1} + \beta_3 \sigma_{i,t-120:t-1} + \varepsilon_{i,t}, \quad (27)$$

where $\rho_{i,t}$ represents the futures-spot deviation, $\beta_{0,i}$ is the currency fixed effect, t is the number of days since the beginning of the sample, $R_{i,t-120:t-1}$ is the annualized spot return to currency i over the past 120 days, and $\sigma_{i,t-120:t-1}$ is the annualized standard deviation of spot return of currency i over the past 120 days. Since we scale the time trend by the number of trading days in a year, the regression coefficient β_1 has an interpretation of the average annual change in the futures-spot deviation.

Table 10 presents the regression results. The regression coefficients on the time trend are negative and highly significant for both ρ and $|\rho|$, indicating that the futures-spot gap has been decreasing both in level and in magnitude over time on average. Throughout our sample, on average, $|\rho|$ shrinks about 11% per year, which accumulates to over 44% over the whole sample. This indicates that the perpetual futures market is becoming more efficient over time.

The coefficients on past returns are positive and highly significant for both ρ and $|\rho|$. This suggests that when past returns are high, perpetual futures exhibit a more positive deviation against the spot. Mapping this back to our second hypothesis, when past returns are high, the demand for futures relative to the spot is also likely to be high. Consequently, even after arbitrageurs accommodate the demand outside of trading costs, the residual demand still manifests itself through the perpetual-spot deviation.

This observation on limits to arbitrage is related to [Makarov and Schoar \(2020\)](#) which finds that crypto price deviations across international exchanges tend to comove with one

Dependent Variable:	ρ			$ \rho $		
Time	-0.20*** (-4.84)	-0.13*** (-3.59)	-0.23*** (-6.78)	-0.22*** (-6.09)	-0.16*** (-5.04)	-0.23*** (-8.22)
Ret		0.10*** (4.45)	0.17*** (10.79)		0.08*** (4.75)	0.13*** (7.67)
Vol			-0.03*** (-6.59)			-0.02*** (-4.52)
Currency FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.07	0.17	0.22	0.12	0.21	0.25
N	7236	7236	7236	7236	7236	7236

Table 10 Regression of the no-arbitrage deviations against explanatory variables

We present regression results of ρ and $|\rho|$ on a time trend (Time) measured in units of years, past 120 days' annualized returns (Ret), and volatility (Vol) in the spot market. We add a currency fixed effect to the regression and consider three specifications: (1) regressing $\rho/|\rho|$ on the time trend, (2) regressing $\rho/|\rho|$ on past returns, and (3) regressing $\rho/|\rho|$ on past returns and volatility. The observations are at a daily frequency. The sample period spans from January 8, 2020 to March 11, 2024, totaling 1,495 days. The table also reports the R^2 values for each regression model. We report Newey-West t-statistics in parentheses with a lag of 120 days. * $p < .1$, ** $p < .05$, *** $p < .01$.

another. The driving force behind this comovement is investors' buying pressure across various countries, with cross-country capital controls serving as the primary impediment to arbitrage capital. In the context of perpetual futures, end-user demand contributes to the comovement of different cryptocurrencies. The limits to arbitrage predominantly manifest in the form of trading frictions.

[Liu and Tsyvinski \(2021\)](#) document a significant time-series momentum pattern in the crypto market. Given that positive past returns lead to a positive gap between futures and spot prices, it is worth examining further whether the time-series momentum phenomenon is driven by margin trading and price pressure from the perpetual futures market. We leave this as an open question for future research.

We find that volatility does not significantly covary with $|\rho|$. This suggests that time-varying funding constraints of arbitrageurs do not drive the comovement in futures-spot deviations across different cryptocurrencies.

In summary, a prerequisite for the gap between futures and spot prices to occur is the existence of trading costs for arbitrageurs. Arbitrage trading will accommodate all the demand in the futures until the price deviation is within a trading cost bound. Within the bound, the relative demand of futures compared to the spot will still manifest itself in the deviation of futures prices from the spot. Due to the comovement of the time-varying relative demand, we observe a significant common factor in crypto future-spot deviations.

6 Conclusion

Perpetual futures play an important role in today's crypto markets and could potentially be adopted in non-crypto markets in the future. Understanding the fundamental mechanism of this financial derivative is a crucial first step for understanding speculation and hedging dynamics in this fast-evolving area. We provide a comprehensive analysis of the arbitrage and funding rate payment mechanisms that underpin perpetual futures.

In an ideal, frictionless world, we show that arbitrageurs would trade perpetual futures in such a way that a constant proportional relationship would hold between the futures price and the spot price. In the presence of trading costs, the deviation of the futures price from the spot would lie within a bound.

Motivated by our theory, we empirically examine the comovement of the futures-spot spread across different cryptocurrencies and implement a theory-motivated arbitrage strategy. We find that this simple strategy yields substantial Sharpe ratios across various trading cost scenarios. The evidence supports our theoretical argument that perpetual futures-spot spreads exceeding trading costs represent a random-maturity arbitrage opportunity.

Finally, we provide an explanation for the common comovement in futures-spot spreads across different crypto-currencies: arbitrageurs can only accommodate market

demand if the price deviation exceeds trading costs. As a result, the overall sentiment in the futures market relative to the spot market is reflected in the spread. Our empirical findings suggest that past return momentum can account for a significant portion of the time-series variation in the futures-spot spread.

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Appendix

A Additional Figures

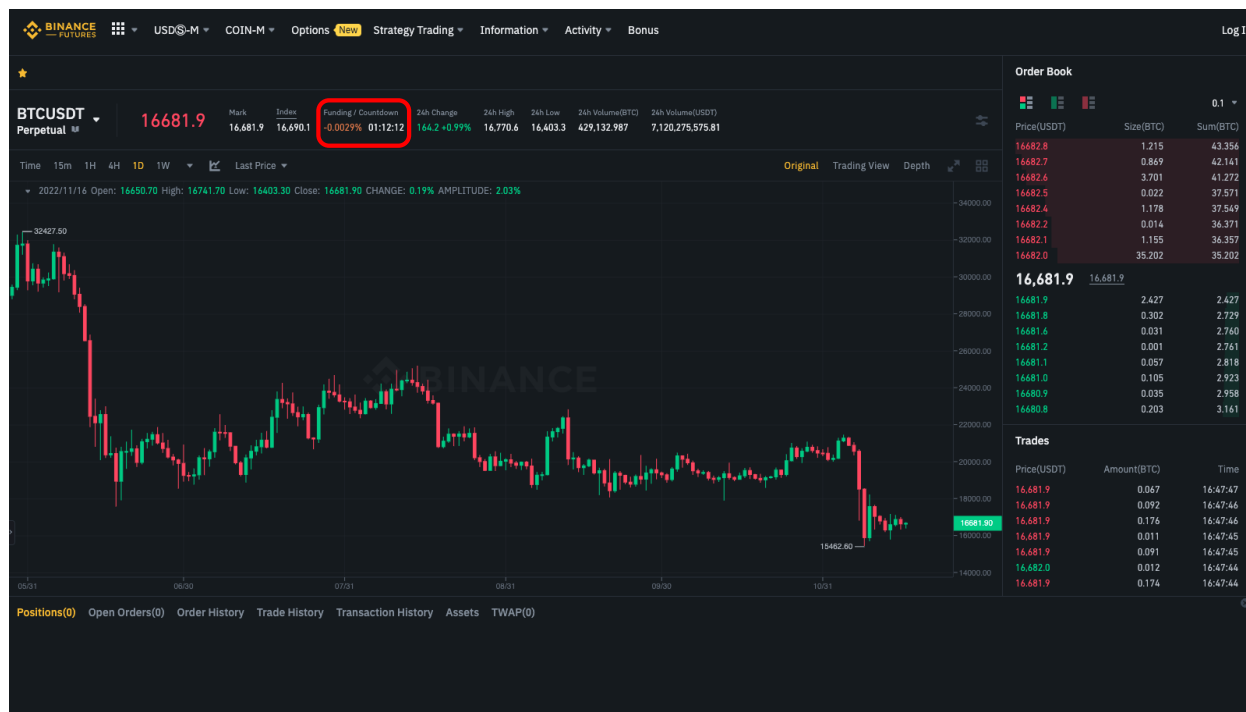


Figure A.1 Trading view of BTC perpetual futures on Binance

This figure presents the perpetual futures trading view on Binance. The key information includes the futures price (Mark), spot price (Index), real-time funding rate based on the rolling average of the past 8 hours' observations, and countdown toward funding rate payment (Funding / Countdown) (illustrated with a red rectangle). In this example, futures are trading at a lower price than the spot. The funding rate is negative and is to be paid in 1 hour and 12 minutes.

Fee Rate

Spot Trading

Margin Borrow Interest

USD\$-M Futures Trading

COIN-M Futures Trading

Options Trading

Swap Farming

P2P

Level	30d Trade Volume (BUSD)	and/or	BNB Balance	Maker / Taker	Maker / Taker BNB 25% off
Regular User	< 1,000,000 BUSD	or	≥ 0 BNB	0.1000% / 0.1000%	0.0750% / 0.0750%
VIP 1	≥ 1,000,000 BUSD	and	≥ 25 BNB	0.0900% / 0.1000%	0.0675% / 0.0750%
VIP 2	≥ 5,000,000 BUSD	and	≥ 100 BNB	0.0800% / 0.1000%	0.0600% / 0.0750%
VIP 3	≥ 20,000,000 BUSD	and	≥ 250 BNB	0.0700% / 0.1000%	0.0525% / 0.0750%
VIP 4 🌟	≥ 100,000,000 BUSD	and	≥ 500 BNB	0.0200% / 0.0400% 0.0700% / 0.0900%	0.0150% / 0.0300% 0.0525% / 0.0675%
VIP 5 🌟	≥ 150,000,000 BUSD	and	≥ 1,000 BNB	0.0200% / 0.0400% 0.0600% / 0.0800%	0.0150% / 0.0300% 0.0450% / 0.0600%
VIP 6 🌟	≥ 400,000,000 BUSD	and	≥ 1,750 BNB	0.0200% / 0.0400% 0.0500% / 0.0700%	0.0150% / 0.0300% 0.0375% / 0.0525%
VIP 7 🌟	≥ 800,000,000 BUSD	and	≥ 3,000 BNB	0.0200% / 0.0400% 0.0400% / 0.0600%	0.0150% / 0.0300% 0.0300% / 0.0450%
VIP 8 🌟	≥ 2,000,000,000 BUSD	and	≥ 4,500 BNB	0.0200% / 0.0400% 0.0300% / 0.0500%	0.0150% / 0.0300% 0.0225% / 0.0375%
VIP 9	≥ 4,000,000,000 BUSD	and	≥ 5,500 BNB	0.0200% / 0.0400%	0.0150% / 0.0300%

• “Taker” is an order that trades at a market price, “Maker” is an order that trades at a limited price. [Learn more](#)

• VIP trade volume levels are measured on the basis of the spot trading volume, or whether the futures trading volume meets the standard (Futures trading volume includes USD\$-M futures and COIN-M futures).

• Refer friends to earn trading fees 20% kickback. [Learn more](#)

Figure A.2 Trading costs tiers for the spot market

This figure presents the trading cost tiers for the crypto spot market from Binance: <https://www.binance.com/en/fee/trading>. Our high, medium, and low trading cost specification corresponds to VIP 1, VIP 5, and VIP 8 tiers, as illustrated with red rectangles in the picture. The 30-day trading volume requirements for VIP 1, 5, and 8 are 1 million, 150 million, and 2 billion, respectively, in the spot market. We consider the trading costs for makers as institutions typically trade maker orders. Binance offers a temporary discount for VIP 4-8 to have the same trading cost as VIP 9. We consider the non-discounted trading costs to make the comparison more reliable and fair.

Fee Rate

Spot Trading Margin Borrow Interest **USDS-M Futures Trading** COIN-M Futures Trading Options Trading Swap Farming P2P

Level	30d Trade Volume (BUSD)	and/or	BNB Balance	USDT Maker / Taker	USDT Maker/Taker <u>BNB 10% off</u>	BUSD Maker / Taker	BUSD Maker/Taker <u>BNB 10% off</u>
Regular User	< 15,000,000 BUSD	or	≥ 0 BNB	0.0200%/0.0400%	0.0180%/0.0360%	0.0120%/0.0300%	0.0108%/0.0270%
VIP 1	≥ 15,000,000 BUSD	and	≥ 25 BNB	0.0160%/0.0400%	0.0144%/0.0360%	0.0120%/0.0300%	0.0108%/0.0270%
VIP 2	≥ 50,000,000 BUSD	and	≥ 100 BNB	0.0140%/0.0350%	0.0126%/0.0315%	0.0120%/0.0300%	0.0108%/0.0270%
VIP 3	≥ 100,000,000 BUSD	and	≥ 250 BNB	0.0120%/0.0320%	0.0108%/0.0288%	0.0120%/0.0300%	0.0108%/0.0270%
VIP 4	≥ 600,000,000 BUSD	and	≥ 500 BNB	0.0100%/0.0300%	0.0090%/0.0270%	0.0100%/0.0300%	0.0090%/0.0270%
VIP 5	≥ 1,000,000,000 BUSD	and	≥ 1,000 BNB	0.0080%/0.0270%	0.0072%/0.0243%	-0.0100%/0.0230%	-0.0100%/0.0207%
VIP 6	≥ 2,500,000,000 BUSD	and	≥ 1,750 BNB	0.0060%/0.0250%	0.0054%/0.0225%	-0.0100%/0.0230%	-0.0100%/0.0207%
VIP 7	≥ 5,000,000,000 BUSD	and	≥ 3,000 BNB	0.0040%/0.0220%	0.0036%/0.0198%	-0.0100%/0.0230%	-0.0100%/0.0207%
VIP 8	≥ 12,500,000,000 BUSD	and	≥ 4,500 BNB	0.0020%/0.0200%	0.0018%/0.0180%	-0.0100%/0.0230%	-0.0100%/0.0207%
VIP 9	≥ 25,000,000,000 BUSD	and	≥ 5,500 BNB	0.0000%/0.0170%	0.0000%/0.0153%	-0.0100%/0.0230%	-0.0100%/0.0207%

• "Taker" is an order that trades at a market price, "Maker" is an order that trades at a limited price. [Learn more](#)

• VIP trade volume levels are measured on the basis of the spot trading volume, or whether the futures trading volume meets the standard (Futures trading volume includes USDS-M futures and COIN-M futures).

• Refer friends to earn trading fees 20% kickback. [Learn more](#)

Figure A.3 Trading costs tiers for the perp market

This figure presents the trading cost tiers for the crypto perpetual market from Binance: <https://www.binance.com/en/fee/trading>. Our high, medium, and low trading cost specification corresponds to VIP 1, VIP 5, and VIP 8 tiers, as illustrated with red rectangles in the picture. The 30-day trading volume requirements for VIP 1, 5, and 8 are 15 million, 1 billion, and 12.5 billion, respectively, in the perpetual market. We consider the trading costs for makers as institutions typically trade maker orders.

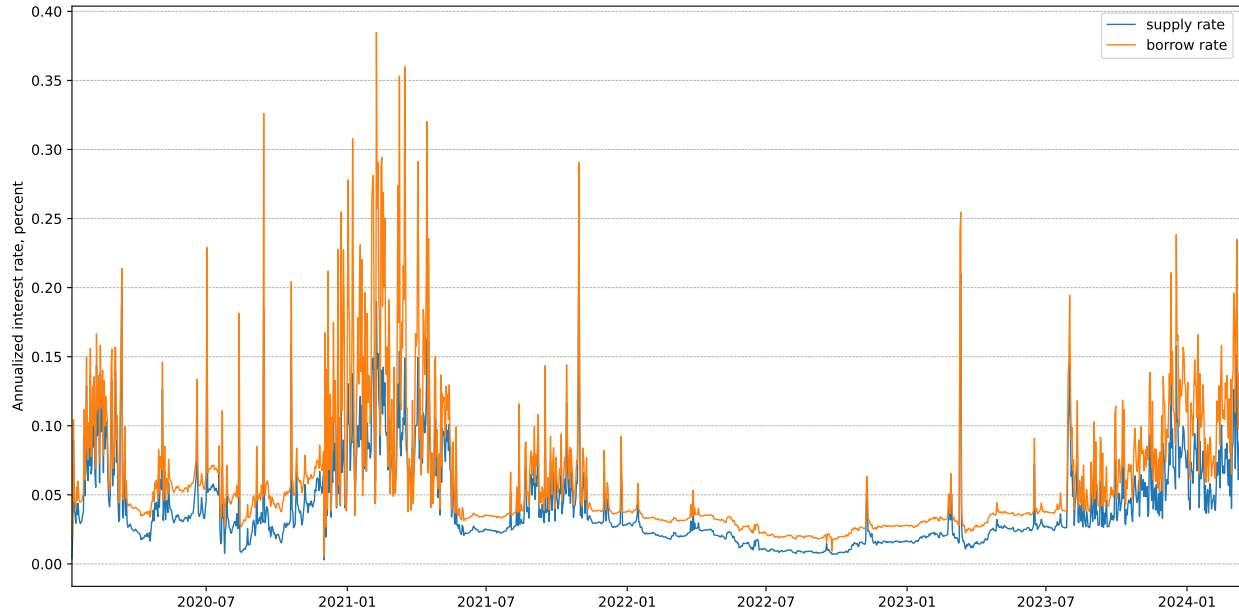


Figure A.4 Annualized interest rate from Aave

This figure presents the daily supply and borrowing rate from Aave. We consider three stablecoins: USDT, USDC, and DAI. Each day, we take the average interest rate of the three crypto stablecoins to get the proxy for the risk-free funding rate of the arbitrageurs. The averaging removes the idiosyncratic noise in borrowing and supply in individual stablecoins. The sample period is from 2020-01-08 to 2024-03-11.

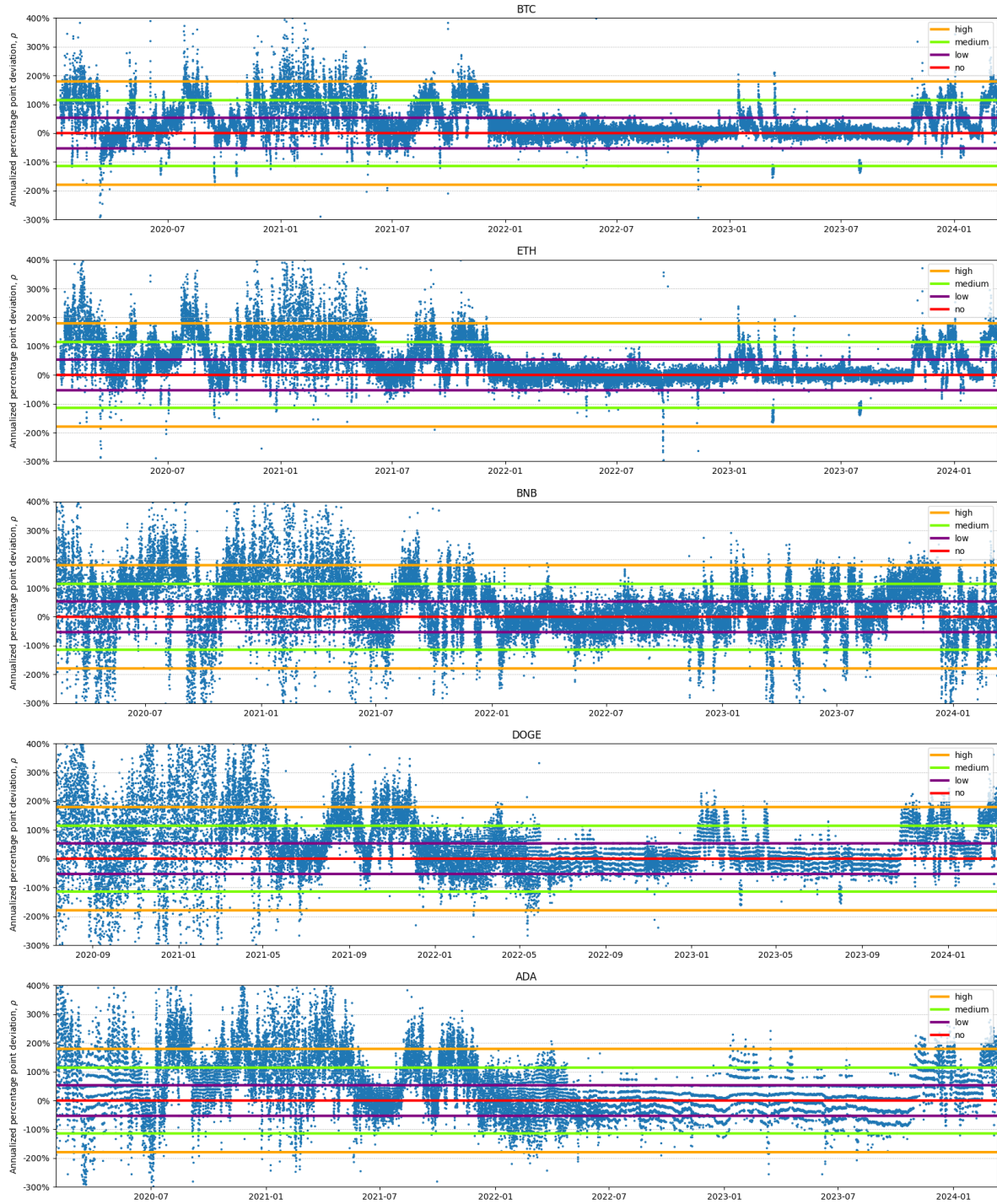


Figure A.5 Trading Strategy Visualization: Random Maturity Arbitrage

This figure presents the random-maturity arbitrage strategy motivated by the theory. The orange, green, and purple lines correspond to the open-position threshold under high, medium, and low trading costs. The red line represents the close-position threshold.

B Funding Rate Calculation on Binance

In this section, we discuss in more detail the funding rate mechanism on the Binance platform based on the methodology given on the Binance website.¹¹

There are five general steps in obtaining the funding rate. The first step is to find the Impact Bid/Ask Price Series in a given funding period by analyzing the order book and determining the average fill price to execute a certain notional value (Impact Margin Notional) on the bid and ask sides.

The impact margin notional is defined as the notional available to trade with a margin of 200 USD. So it can be calculated using the following equation:

$$\text{IMN} = 200 / \text{Initial margin rate at maximum leverage level.}$$

Given the definition of impact margin notional, the impact bid/ask price is defined as the average trade price after a market sell/buy order with the total quoted notional quantity given by the IMN.

Given impact bid and ask prices (denoted as IBP and IAP, respectively), the second step is to calculate the premium index (denoted as P) series every 5 seconds using the following equation:

$$P = [\text{Max}(0, \text{IBP} - \text{Price Index}) - \text{Max}(0, \text{Price Index} - \text{IAP})] / \text{Price Index},$$

where 'Price Index' is the weighted average spot price of the underlying asset listed on major spot exchanges.

The rationale behind this formula is to capture the relative difference between the perpetual contract's price and the underlying asset's price, taking into account the actual market liquidity and depth represented by the IBP and IAP.

The third step is to calculate the time-weighted average premium index over the funding period (8 hours for Binance) using the premium index series:

$$\bar{P} = \sum_{t=1}^T w_t P_t / \sum_{t=1}^T w_t,$$

where P_t denotes the t -th 5-second in the 8-hour funding period, $w_t = t$ is the weight for

¹¹<https://www.binance.com/en/support/faq/360033525031>

each observation, and T is the total number of 5-second intervals within the 8-hour period. For Binance, $T = 12 \times 6 \times 8 = 5760$. With this weighting scheme, the observations closer to the funding rate payment receive larger weights. The weights decrease linearly in time over the 8-hour interval.

Given the average premium index, the fourth step is to calculate the funding rate using the equation below:

$$fr = \bar{P} + \text{clamp}(\iota - \bar{P}, -0.05\%, 0.05\%),$$

where fr is the funding rate, γ is the interest component set by Binance, and

$$\text{clamp}(x, \min, \max) = \begin{cases} \min, & \text{if } x < \min \\ x, & \text{if } \min \leq x \leq \max \\ \max, & \text{if } x > \max \end{cases}$$

Empirically, for Binance futures, the interest rate r is fixed at 0.01% per 8 hours for most perpetual futures. Figure 3 plots the funding rate as a function of the premium index. As can be seen, when the premium index is close to the interest rate 0.01%, the funding is fixed at the interest rate.

This design allows the funding rate to be fixed to a constant positive rate under normal market conditions. When markets are relatively stable and the perpetual contract price is trading close to the index price, the premium index will be small. The clamp ensures that under these normal conditions, the funding rate matches the pre-defined interest rate set by the exchange. This makes funding rate payments more predictable.

Note that this still allows larger funding rates when there are significant deviations. If the premium index differs from the interest rate by more than 0.05% in either direction, the clamp will not cap the funding rate. Larger premiums or discounts in the contract price will flow through to correspondingly larger funding payments to incentivize price convergence.

Last but not least, the fifth step is to apply a Capped Funding Rate based on the maximum leverage of the contract. For contracts with a maximum leverage of 30x or above, the cap and floor are determined by the Maintenance Margin Ratio. For contracts with a maximum leverage of 25x or below, the cap and floor are fixed at $\pm 3\%$. For contracts with higher leverage, the cap and floor can expand based on maximum leverage.

For this design, extremely high funding rates can lead to significant losses for traders holding positions, especially those with high leverage. By capping the funding rate,

Binance limits the potential downside from holding a position during volatile periods. This can help prevent accounts from being liquidated due to surging funding costs.

C Data-driven Arbitrage Strategy

In the main text of our paper, we demonstrate the profitability of a simple theory-motivated trading strategy. The trading threshold can also be potentially further improved using a data-driven approach. In this part, we implement a data-driven two-threshold arbitrage trading strategy in the perpetual futures market to verify our proposed theory-motivated strategy is optimal.

The data-driven strategy can be characterized with a tuple of two thresholds: (u, l) , where u denotes the upper bar and l denotes the lower bar, $u > l$. When $\rho > u$, we long the spot and short the futures. When $-l < \rho < l$, we close the position. When $\rho < -u$, we long the futures and short the spot. Figure C.1 presents an illustration of the strategy for Bitcoin with the trading thresholds estimated using real-time data.

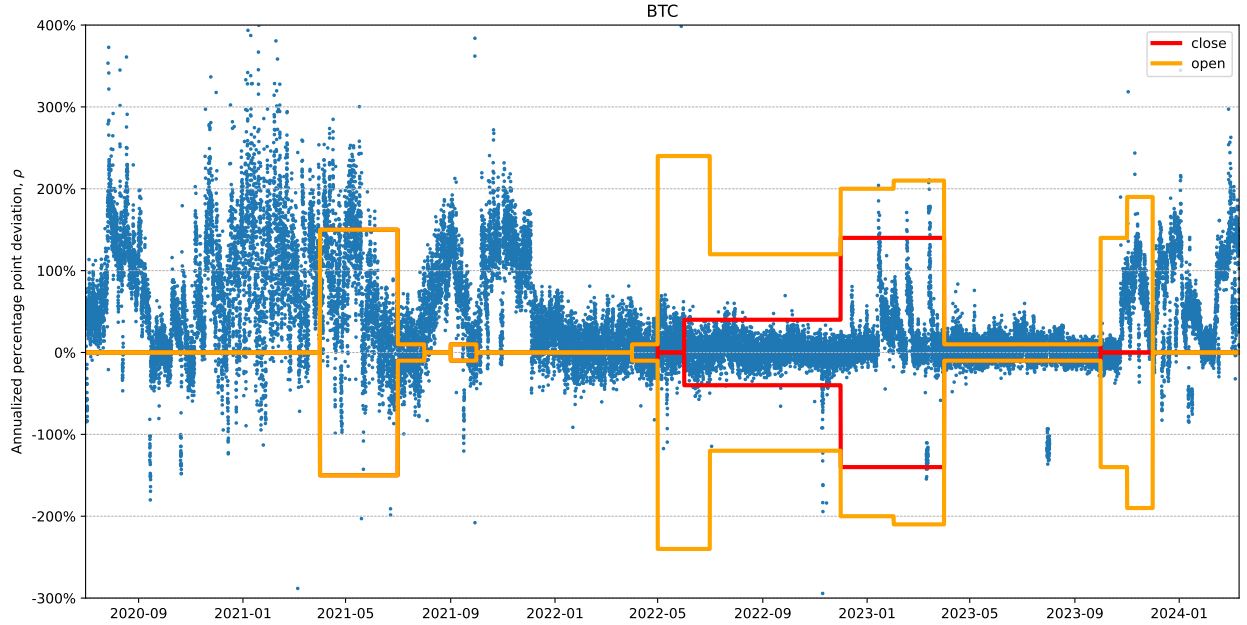


Figure C.1 Data-driven Random Maturity Arbitrage Strategy for Bitcoin

Deviation from no-arbitrage price (ρ) and trading rules of the data-driven random-maturity arbitrage strategy we implement for BTC. Each blue dot in the figure represents the deviation of futures from the no-arbitrage price calculated using Equation 21 at hourly frequency. The number is annualized for ease of interpretation. The orange line is the open-position threshold, and the red line is the close-position threshold.

To determine the optimal (u, l) , at the beginning of each month, we calculate the returns of the two-threshold trading strategy based on the past six months' data on a grid of parameters. From Table 3, we find the theory implied bounds for price deviation across

all trading costs specifications lie within -200% to $+200\%$. So we choose the grid as increasing from 0% to 250% with an incremental step of 10% for u and l ($u \geq l$). In total, there are 351 model specifications. We choose the model that delivers the highest Sharpe Ratio in the validation sample of the past six months.

We present the visualization of all trading strategies under zero to high trading costs in appendix Figure C.2 to Figure C.5. Under no-trading costs, the strategy chooses a low open-position threshold, and the open- and close-position thresholds coincide with each other. Since there is no trading cost, the strategy no longer needs to wait for price convergence to avoid high turnover. On the contrary, with high trading costs, the strategy chooses a larger open-position threshold and a smaller close-position threshold. This reflects the automatic adjustment from the algorithm for trading costs and turnovers.

Table C.1 presents the performance statistics for this strategy under no trading costs over time. Table C.2 further presents the overall performance of the data-driven strategy across different trading costs. We find the arbitrage trading strategy has a high Sharpe ratio similar and comparable to our theory-motivated ones. This suggests our theory-motivated strategy is close to optimal.

Similar to the theory-motivated case, we decompose the performance of the trading strategy into: (1) the price convergence and (2) the funding rate payment. The results are presented in Table C.3. Similar to the theory-motivated strategy, we find that the return from price convergence accounts for a larger proportion of the total profit.

Overall, our analysis of the data-driven trading strategy validates the optimality of our theory-driven approach, underscoring the value of our theoretical benchmark. Notably, introducing additional flexible adjustments does not significantly improve the performance of the arbitrage strategy.

		2020	2021	2022	2023	2024	All
BTC	SR	5.14	10.86	20.28	12.80	11.84	8.43
	Return	35.93	53.76	21.00	21.93	21.44	32.18
	Volatility	6.99	4.95	1.04	1.71	1.81	3.82
	MaxDD	-3.82	-2.00	-0.12	-0.12	-0.17	-3.82
	Active %	100.00	76.42	30.00	32.71	100.00	56.47
	OtC time	10.74	8.66	3.26	2.46	11.44	5.54
ETH	N	4,416	8,760	8,760	8,760	1,682	32,378
	SR	8.60	12.01	20.01	11.90	13.90	11.78
	Return	48.21	61.78	43.04	41.25	25.11	47.40
	Volatility	5.61	5.14	2.15	3.47	1.81	4.02
	MaxDD	-1.09	-1.43	-0.67	-0.29	-0.16	-1.43
	Active %	63.65	48.58	50.09	86.77	100.00	64.05
BNB	OtC time	11.52	5.89	2.55	3.59	8.54	4.15
	N	4,416	8,760	8,760	8,760	1,682	32,378
	SR	18.67	15.51	34.53	21.26	24.05	19.05
	Return	140.41	124.74	108.01	93.76	53.50	109.56
	Volatility	7.52	8.04	3.13	4.41	2.22	5.75
	MaxDD	-0.61	-1.41	-0.83	-0.61	-0.16	-1.41
DOGE	Active %	45.26	56.69	57.45	82.92	19.56	60.86
	OtC time	3.87	5.68	2.44	4.72	2.25	3.81
	N	3,672	8,760	8,760	8,760	1,682	31,634
	SR	-	13.64	44.30	20.96	6.92	14.51
	Return	-	208.60	159.41	41.62	13.23	129.13
	Volatility	-	15.30	3.60	1.99	1.91	8.90
ADA	MaxDD	-	-3.82	-0.29	-0.21	-0.29	-3.82
	Active %	-	80.17	55.59	40.55	16.41	56.22
	OtC time	-	6.00	1.92	2.82	18.40	3.15
	N	-	8,760	8,760	8,760	1,682	27,962
	SR	13.53	12.34	35.09	26.92	13.03	16.15
	Return	87.73	118.98	122.89	68.76	42.29	98.20
	Volatility	6.48	9.64	3.50	2.55	3.25	6.08
	MaxDD	-0.89	-5.69	-0.22	-0.50	-0.18	-5.69
	Active %	48.62	60.99	33.00	33.41	100.00	46.30
	OtC time	5.73	5.24	2.12	2.45	13.90	3.68
	N	4,416	8,760	8,760	8,760	1,682	32,378

Table C.1 Data-driven Random Maturity Arbitrage Strategy over Time

This table presents the Sharpe ratios, annualized returns (%), standard deviations (%), maximum drawdowns (%), active percentages (%), and the average open-to-close time (hours) of the data-driven random maturity arbitrage strategies for five different cryptocurrencies. The returns reported are excess returns from the random maturity arbitrage strategies.

		Fee tiers			
		No	Low	Medium	High
BTC	SR	8.43	3.71	3.49	3.18
	Return	32.18	13.18	12.01	10.88
	Volatility	3.82	3.55	3.44	3.42
	MaxDD	-3.82	-3.82	-2.00	-2.00
	α	32.69	12.50	11.64	10.22
	t_α	10.02	5.31	5.13	4.88
	Active %	56.47	26.38	24.79	23.55
	OtC time	5.54	43.36	89.17	95.33
ETH	SR	11.78	5.88	4.50	4.22
	Return	47.40	21.86	17.17	16.09
	Volatility	4.02	3.72	3.81	3.81
	MaxDD	-1.43	-1.47	-2.62	-2.66
	α	49.84	20.56	16.39	14.91
	t_α	10.79	5.72	5.28	5.07
	Active %	64.05	22.60	23.95	24.24
	OtC time	4.15	16.01	69.85	89.17
BNB	SR	19.05	12.75	8.81	7.19
	Return	109.56	66.86	44.35	36.76
	Volatility	5.75	5.24	5.03	5.11
	MaxDD	-1.41	-1.07	-1.12	-1.42
	α	113.78	53.44	35.05	26.51
	t_α	12.61	6.91	5.30	4.46
	Active %	60.86	41.44	24.69	21.51
	OtC time	3.81	6.70	13.10	18.10
DOGE	SR	14.51	9.19	6.32	4.28
	Return	129.13	80.53	52.43	37.25
	Volatility	8.90	8.76	8.30	8.71
	MaxDD	-3.82	-3.87	-3.92	-5.61
	α	155.82	81.40	43.65	30.03
	t_α	7.26	4.56	3.14	2.76
	Active %	56.22	47.00	26.97	22.88
	OtC time	3.15	6.20	10.22	22.85
ADA	SR	16.15	12.08	6.44	3.86
	Return	98.20	72.14	35.83	21.76
	Volatility	6.08	5.97	5.56	5.64
	MaxDD	-5.69	-5.67	-5.69	-5.58
	α	109.86	65.07	27.92	18.26
	t_α	9.55	9.78	7.09	5.22
	Active %	46.30	50.95	29.20	21.19
	OtC time	3.68	5.40	12.81	40.13

Table C.2 Data-driven Random Maturity Arbitrage across Different Trading Costs

Performance of data-driven random maturity arbitrage strategy under different trading cost tiers. The fees for spot (futures) are 2.25 (0.18) bps, 4.5 (0.72) bps, and 6.75 (1.44) bps for the low, medium, and high trading cost levels. Statistics reported are the annualized Sharpe ratio, return (%), volatility (%), max drawdown (%), alpha (%), t-stat of the alpha, proportion of time the strategy is active (%), and average open-to-close (OtC) position duration in hours.

		2020	2021	2022	2023	2024	All
BTC	Return	35.91	53.76	21.00	21.93	21.44	32.17
	Price	17.53	28.50	19.25	19.28	3.30	20.70
	Funding	18.38	25.27	1.75	2.65	18.15	11.48
ETH	Return	48.19	61.78	43.04	41.25	25.11	47.40
	Price	25.20	42.92	39.62	36.37	7.54	36.00
	Funding	22.99	18.87	3.43	4.88	17.57	11.40
BNB	Return	140.45	124.74	108.01	93.76	53.50	109.56
	Price	114.53	100.82	99.69	81.75	45.62	93.88
	Funding	25.92	23.92	8.32	12.01	7.88	15.68
DOGE	Return	-	208.40	159.41	41.62	13.23	129.06
	Price	-	167.82	157.15	40.50	3.13	114.68
	Funding	-	40.59	2.26	1.12	10.10	14.38
ADA	Return	87.49	118.98	122.89	68.76	42.29	98.17
	Price	59.58	89.46	121.51	67.21	19.16	84.39
	Funding	27.91	29.52	1.37	1.54	23.13	13.79

Table C.3 Data-driven Strategy: Price Convergence vs Funding Rate Payment

This table decomposes the portfolio return into the part due to price convergence and the part due to funding rate payment.

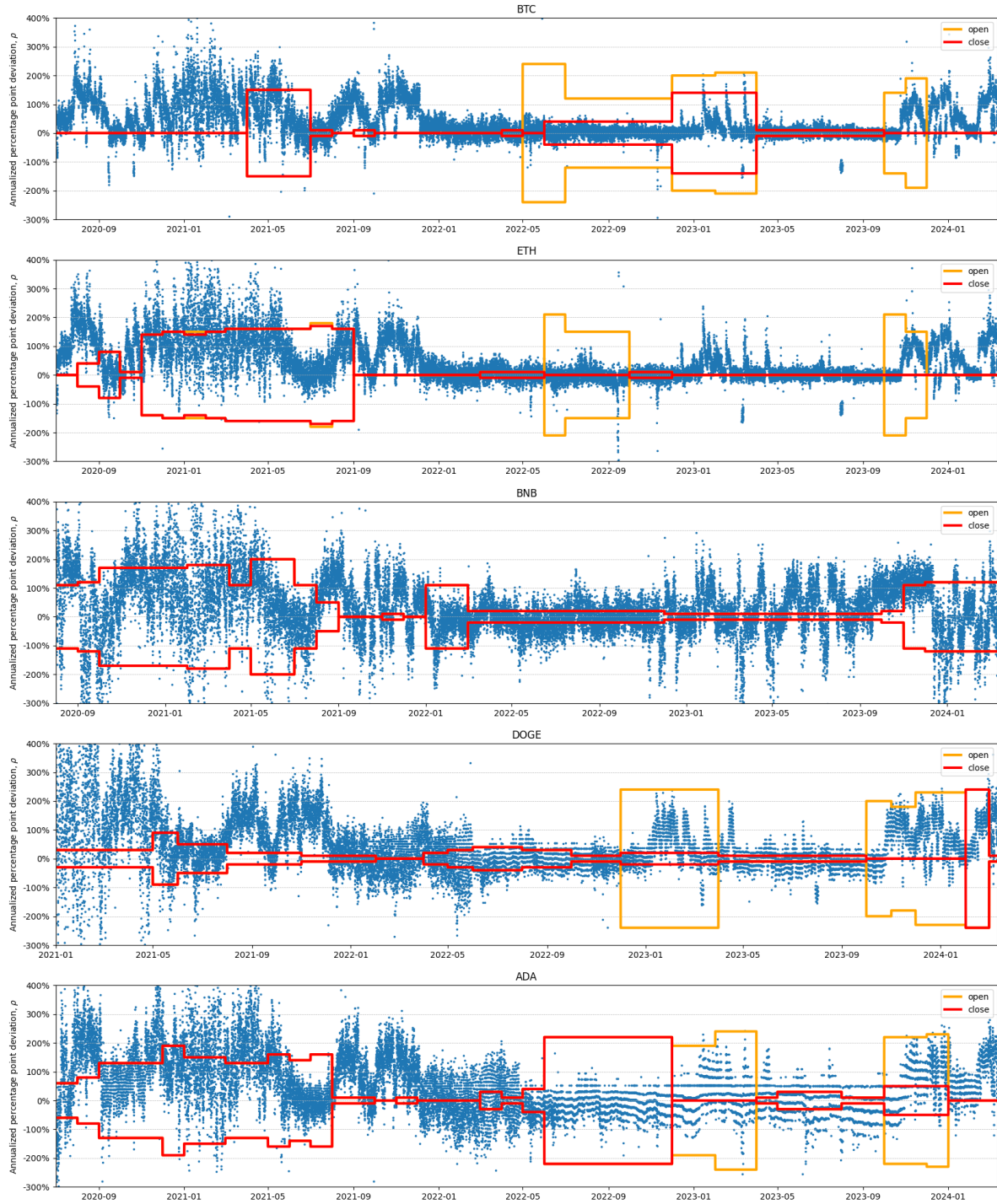


Figure C.2 Data-driven Trading Strategy Visualization: No Trading Costs

This figure presents the real-time trading thresholds under no trading costs. The orange line is the open-position threshold, and the red line is the close-position threshold. The trading thresholds are determined based on the adjusted SR from the past six months.

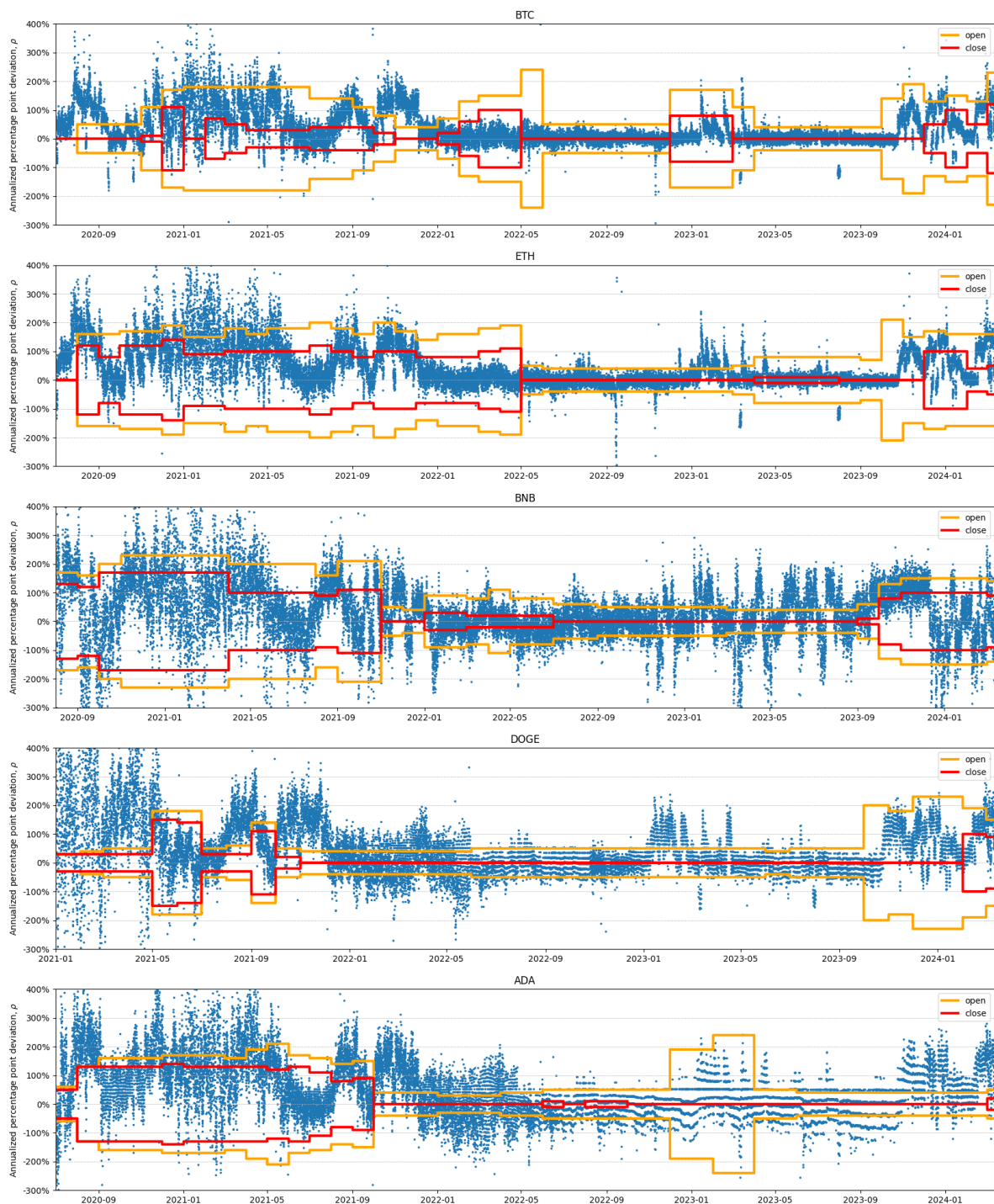


Figure C.3 Data-driven Trading Strategy Visualization: Low Trading Costs

This figure presents the real-time trading thresholds under low trading costs (2.25 bps for spot trading and 0.18 bp for futures trading). The orange line is the open-position threshold, and the red line is the close-position threshold. The trading thresholds are determined based on the adjusted SR from the past six months.

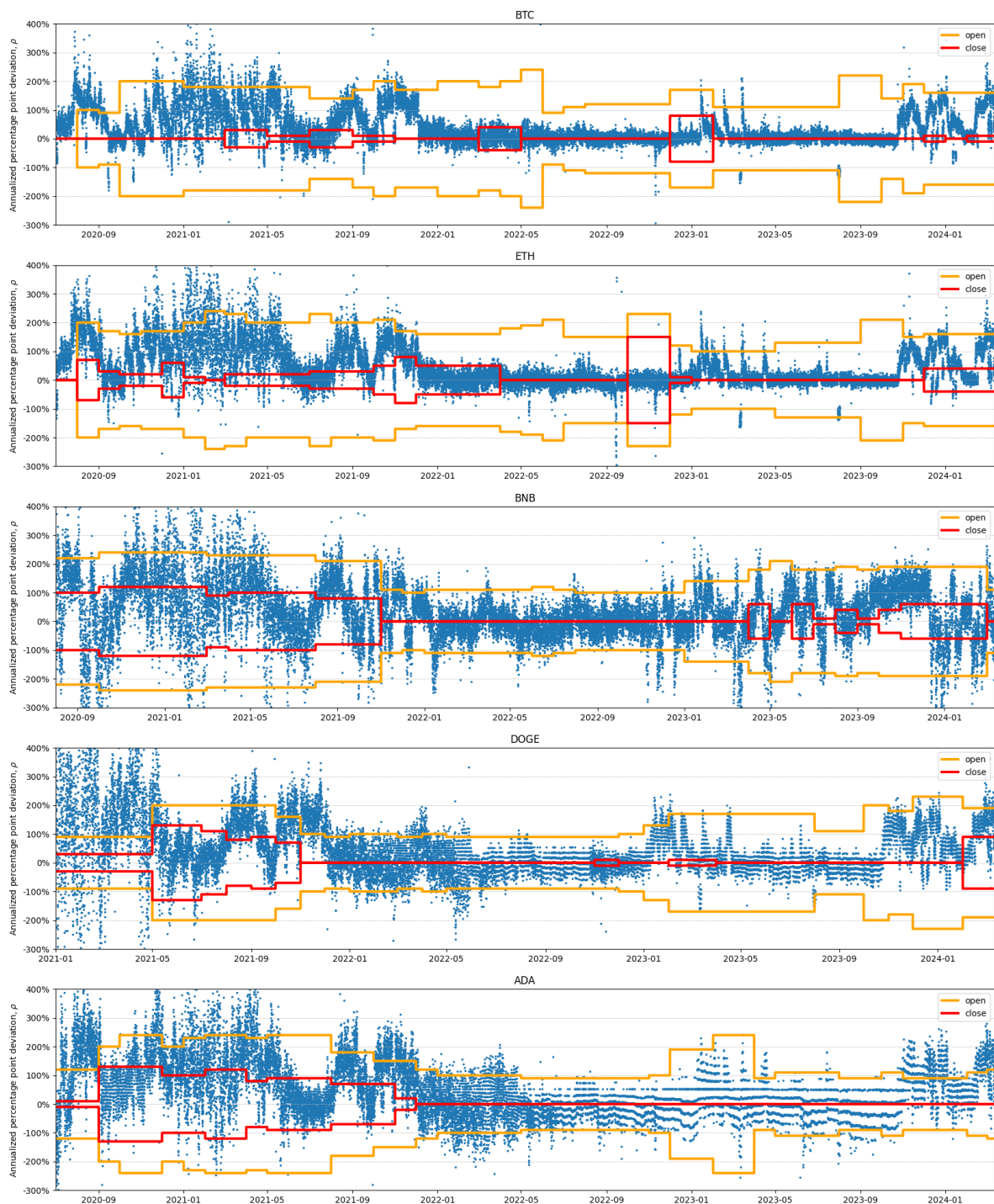


Figure C.4 Data-driven Trading Strategy Visualization: Medium Trading Costs

This figure presents the real-time trading thresholds under medium trading costs (4.5 bps for spot trading and 0.72 bp for futures trading). The orange line is the open-position threshold, and the red line is the close-position threshold. The trading thresholds are determined based on the adjusted SR from the past six months.

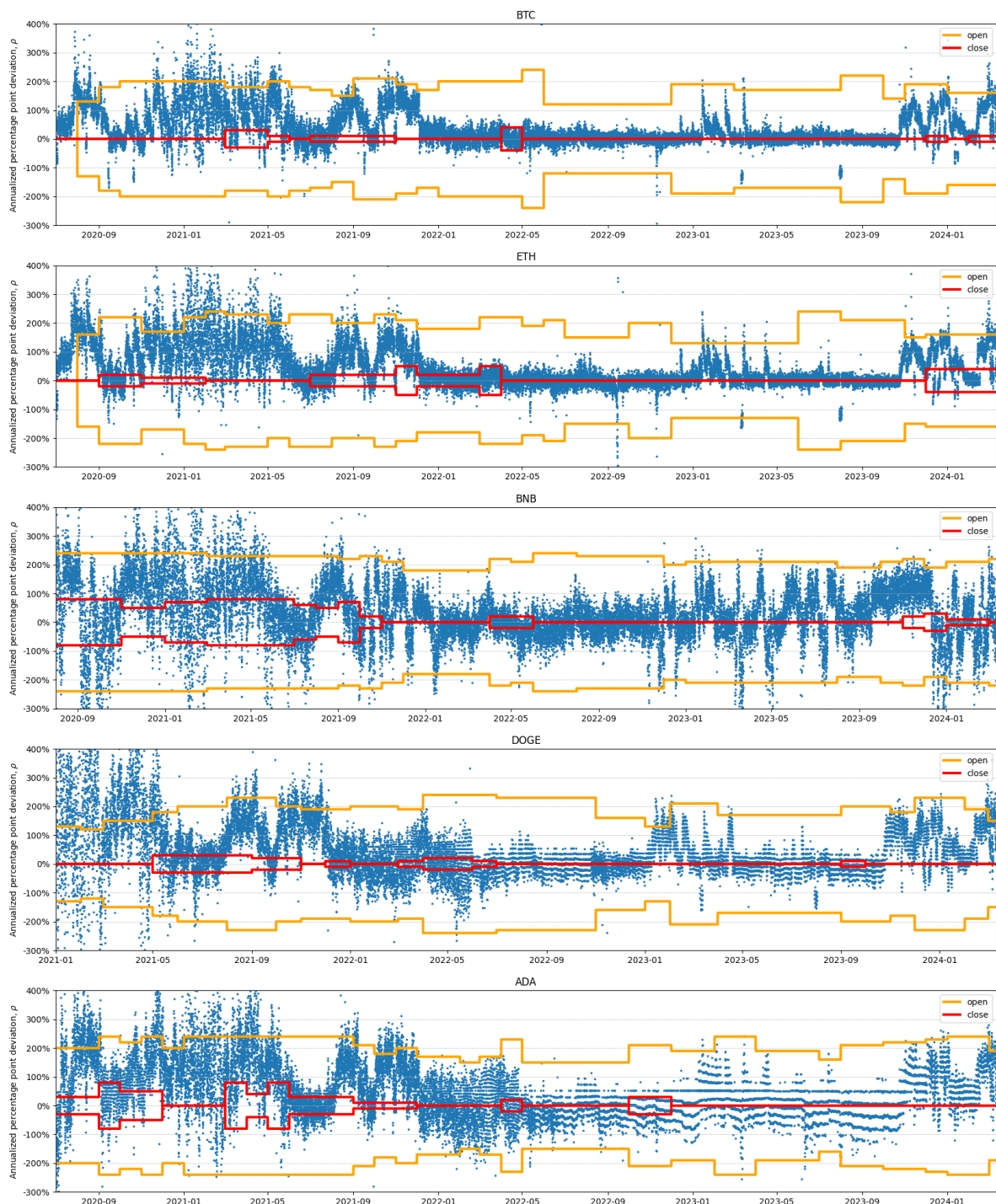


Figure C.5 Data-driven Trading Strategy Visualization: High Trading Costs

This figure presents the real-time trading thresholds under high trading costs (6.75 bps for spot trading and 1.44 bps for futures trading). The orange line is the open-position threshold, and the red line is the close-position threshold. The trading thresholds are determined based on the adjusted SR from the past six months.

D Robustness to Alternative Risk-free Rate Measures

We show robustness in the random-maturity arbitrage strategies' performance by considering alternative risk-free rates and present the results in Table D.1. We conduct our analysis using daily T-bill rates sourced from Kenneth French's data library (https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/f-f_factors.html) and margin interest rates from Binance (https://binance-docs.github.io/apidocs/spot/en/#query-margin-interest-rate-history-user_data). We find that the performance of the trading strategy is robust to these alternative risk-free rates and across different trading costs tiers.

		No Trading Cost		High Trading Cost	
		FF	Binance	FF	Binance
BTC	SR	11.65	11.73	3.79	2.88
	Return	52.60	52.91	15.57	10.45
	Volatility	4.52	4.51	4.11	3.63
	MaxDD	-3.82	-3.82	-3.82	-2.63
	α	59.60	54.40	14.28	12.11
	t_α	18.59	15.65	5.91	7.03
	Active %	100.00	100.00	31.30	19.37
	OtC time	5.20	5.12	200.86	97.05
ETH	SR	12.35	14.17	4.93	4.47
	Return	60.83	69.50	21.23	15.98
	Volatility	4.92	4.90	4.31	3.58
	MaxDD	-2.62	-2.57	-2.62	-1.59
	α	70.74	75.07	21.23	17.07
	t_α	18.28	12.65	6.17	5.98
	Active %	100.00	100.00	35.27	22.25
	OtC time	5.38	4.89	189.74	59.85
BNB	SR	16.94	19.40	6.63	7.44
	Return	129.42	147.17	43.13	45.51
	Volatility	7.64	7.59	6.50	6.11
	MaxDD	-1.97	-1.96	-2.09	-2.08
	α	142.69	154.89	36.18	38.04
	t_α	15.19	13.78	6.50	6.25
	Active %	100.00	100.00	41.32	35.78
	OtC time	5.79	5.03	35.45	30.05
DOGE	SR	14.45	14.87	4.91	5.33
	Return	175.82	180.22	55.19	55.63
	Volatility	12.17	12.12	11.23	10.43
	MaxDD	-13.90	-13.87	-13.90	-10.08
	α	218.01	205.74	45.21	48.22
	t_α	11.30	10.24	4.80	5.38
	Active %	100.00	100.00	33.28	27.55
	OtC time	4.29	4.13	19.67	13.88
ADA	SR	19.50	19.45	4.28	3.87
	Return	153.28	152.70	28.75	24.91
	Volatility	7.86	7.85	6.72	6.43
	MaxDD	-5.68	-5.65	-5.68	-5.58
	α	175.95	151.87	25.42	22.08
	t_α	19.85	18.03	6.49	6.42
	Active %	100.00	100.00	33.70	23.79
	OtC time	4.46	4.23	56.71	29.25

Table D.1 Robustness Check: Alternative Risk-free Rates

Performance under alternative risk-free rates. ‘FF’ represents the risk-free rate from Ken French’s data library. ‘Binance’ represents the margin borrowing rate from Binance. We report performance under no trading costs and high trading costs. Statistics reported are the annualized Sharpe ratio, return (%), volatility (%), max drawdown (%), alpha (%), t-stat of the alpha, proportion of time the strategy is active (%), and average open-to-close (OtC) position duration in hours.

E Proofs

In this section, we provide proofs to Proposition 2, Corollary 2, and Proposition 3.

Proof to Proposition 2:

Proof. Before proceeding, define $\bar{\lambda} := (1 - \sup \{\Phi^{-1}(r)\})^{-1}$ and $\underline{\lambda} := (1 - \inf \{\Phi^{-1}(r)\})^{-1}$. These quantities correspond to the upper and lower bounds of the interval in Equation (11).

We first show that if the futures price satisfies Equation (11), then there are no random maturity arbitrage opportunities. In this case, consider the two strategies in Table 1. We consider two types of strategies, those where λ in Table 1 is within the range implied by Equation (11) and those where it is outside this range.

Firstly, $\forall \lambda \in [\underline{\lambda}, \bar{\lambda}]$, there exist price paths $F_t = \lambda S_t$ such that Equation (11) is satisfied and the payoff to the arbitrageur is 0. Therefore, neither strategy is random maturity arbitrage.

It remains to show that when λ is outside the range, the long-short strategy is not a random maturity arbitrage. This is the case because such strategies would be exposed to directional risk in the spot or futures market, which is clearly not a random maturity arbitrage strategy. Therefore, there is no random maturity arbitrage opportunity in this market.

Next, we demonstrate that a price deviation from the interval defined in Equation (11) would imply the existence of random maturity arbitrage opportunities. Consider the following two scenarios: (1) $F_0 > \bar{\lambda} S_0$; (2) $F_0 < \underline{\lambda} S_0$.

Scenario 1: When $F_0 > \bar{\lambda} S_0$, define the deviation from fundamental price as:

$$u_s \equiv F_s - \bar{\lambda} S_s. \quad (28)$$

Following the same logic as the proof for Proposition 1, we consider the first strategy in Table 1 of longing $\bar{\lambda}$ spot and shorting 1 futures, whose payoff is identical to Equation (7). Following the same logic, in this case, the proposed strategy is a random maturity arbitrage strategy.

Scenario 2: When $F_0 < \underline{\lambda} S_0$, define the deviation from fundamental price as:

$$u_s \equiv F_s - \underline{\lambda} S_s. \quad (29)$$

Similar to Scenario 2 in Proposition 1, we consider the second strategy in Table 1 of longing 1 futures and shorting λ spot, whose payoff is the same as Equation (9). Following the same logic, in this case, the proposed strategy is a random maturity arbitrage strategy. $\square$

Proof to Corollary 2:

Proof. In this proof, we first demonstrate how the benchmark price in Equation (13) is derived. Next, we establish that if the futures price satisfies this relationship, random-maturity arbitrage opportunities are absent. Finally, we prove that the absence of random-maturity arbitrage opportunities necessitates that the futures price conforms to Equation (13), as any deviation would violate this assumption.

To show how we obtain the formula, start off with the conjecture that in equilibrium, the futures price will be a linear function of the spot: $F_t = \lambda S_t$.

If this conjecture is true, it implies both strategies laid out in Table 1 should always have 0 payoff. We can write down the payoff of the first strategy as:

$$-\int_0^t e^{-rs} d(F_s - \lambda S_s) + \int_0^t e^{-rs} (\Phi F_s - r\lambda S_s) ds = (\Phi - r) \int_0^t e^{-rs} F_s ds = 0, \quad (30)$$

where the first equality follows from the condition $F_t = \lambda S_t, \forall t$. From the second equality, we obtain the condition for pinning down the value of λ : $\Phi = r$, which states that in equilibrium, the funding rate should equal the differences in risk-free rates, i.e.:

$$\kappa \frac{F_s - S_s}{F_s} + \text{clamp}(\iota - \kappa \frac{F_s - S_s}{F_s}, -\gamma, \gamma) = r. \quad (31)$$

Solving the above Equation gives us exactly the relationship between F and S as given in Equation (13).

If Equation (13) always holds, then both strategies in Table 1 have zero payoff forever. Any other long-short strategy with different proportionality would be exposed to directional risk in the spot market. Therefore, this implies that there is an absence of random-maturity arbitrage opportunity across the futures and spot markets.

Next, we demonstrate that any deviation from this benchmark results in random-maturity arbitrage opportunities. To prove this, we separate into two cases: (1) $\iota < r$ and (2) $\iota > r$. We focus on the $\iota < r$ case below and note that a similar argument applies to the $\iota > r$ case.¹²

¹²When $\iota = r$, Proposition 2 implies that the fundamental price for perpetual futures will fall within an interval. Notably, when r is stochastic, the probability of $r = \iota$ is zero. Consequently, we abstract from this

Assuming that $\iota < r$, we are going to show that if there is no arbitrage, then

$$F_t = \frac{\kappa}{\kappa - \gamma - r} S_t. \quad (32)$$

We consider a deviation from this benchmark and show that it implies the existence of random maturity arbitrage opportunities.

Before proceeding, we define the following regions of the funding rate. The clamp region is where the futures price F and the spot price S satisfy the condition: $\kappa \frac{F_s - S_s}{F_s} - \gamma \leq \iota \leq \kappa \frac{F_s - S_s}{F_s} + \gamma$. The upper region is defined as the range where F and S satisfy $\iota < \kappa \frac{F_s - S_s}{F_s} - \gamma$. Conversely, the lower region is where F and S satisfy $\kappa \frac{F_s - S_s}{F_s} + \gamma < \iota$. In this case, we define the deviation from the fundamental price as:

$$u_s \equiv F_s - \frac{\kappa}{\kappa - \gamma - r} S_s. \quad (33)$$

Scenario 1: Firstly, consider the scenario where $F_0 > \lambda S_0$. In this scenario, the premium index falls in the upper region of the Φ function. To arbitrage the deviation, we consider the first strategy in Table 1. Its payoff can be written as:

$$-\int_0^t e^{-rs} du_s + (\kappa - \gamma) \int_0^t e^{-rs} u_s ds = -e^{-rt} u_t + u_0 + (\kappa - \gamma - r) \int_0^t u_s e^{-rs} ds, \quad (34)$$

where the second equality follows from integration by parts. To prevent the arbitrageur from making a profit, the payoff must be nonpositive for all times t :

$$\underbrace{e^{-rt} u_t}_{\text{spread at unwinding}} \geq \underbrace{u_0}_{\text{initial spread}} + \underbrace{(\kappa - \gamma - r) \int_0^t u_s e^{-rs} ds}_{\text{funding payments}}. \quad (35)$$

Consider the first time period when the deviation is positive and let t^* denote the first time it hits zero again. That is, $u_s > 0, s \in (0, \min\{t^* > 0, u_{t^*} = 0\})$.

If $t^* < \infty$, then when $t = t^*$, the LHS of Inequality (35) equals 0 while the RHS of Inequality (35) is larger than 0. Therefore, the payoff at t^* to the arbitrageur is positive. It remains to show that if $t^* \rightarrow \infty$ the payoff to the arbitrageur will also be positive.

Let $\underline{u}_t$ denote a process, implicitly defined by $e^{-rt} \underline{u}_t = u_0 + (\kappa - \gamma - r) \int_0^t u_s e^{-rs} ds$, which provides a lower bound for all processes u_t satisfying Inequality (35). Solving this integral

scenario but acknowledge that a similar argument as in Proposition 2 can be extended to it.

equation, we have:

$$\underline{u}_t = e^{(\kappa - \gamma)t} u_0. \quad (36)$$

This lower bound process explodes towards ∞ at an exponential speed, which would violate the Assumption 2. As a result, if $t^* \rightarrow \infty$, the payoff to the arbitrageur will also be positive.

Therefore, if $F_0 > \lambda S_0$, the proposed strategy is a random maturity arbitrage strategy.

Scenario 2: Next, consider the scenario where $F_0 < \lambda S_0$. In this scenario, it is possible that the premium index falls in the clamp region. Now, consider the second strategy in Table 1. The payoff to this strategy is given by:

$$\begin{aligned} \int_0^t e^{-rs} du_s - \int_0^t e^{-rs} (\Phi F_s - r\lambda S_s) ds &= e^{-rt} u_t - u_0 + r \int_0^t e^{-rs} u_s ds \\ &\quad - \int_0^t e^{-rs} (\Phi F_s - r\lambda S_s) ds, \end{aligned} \quad (37)$$

where the equality follows from integration by parts. By the same argument of *Scenario 1*, the u_s will increase at an exponential speed until the futures and spot prices reach the clamp region. When the futures and spot prices lie in the clamp region, the payoff is given by:

$$\begin{aligned} &e^{-rt} u_t - u_0 + r \int_0^t e^{-rs} (F_s - \lambda S_s) ds - \int_0^t e^{-rs} (1F_s - r\lambda S_s) ds \\ &= e^{-rt} u_t - u_0 - \int_0^t e^{-rs} (1 - r) F_s ds \end{aligned} \quad (38)$$

In order to keep the arbitrageurs away from making a profit, the above expression should satisfy:

$$e^{-rt} u_t \leq u_0 + \int_0^t e^{-rs} (1 - r) F_s ds. \quad (39)$$

Consider again the first time period the deviation is negative and let t^* denote the first time it hits zero again. That is $u_s < 0, s \in (0, \min\{t^* > 0, u_{t^*} = 0\})$.

If $t^* < \infty$, then when $t = t^*$, the LHS of Inequality (39) is 0 while the RHS is negative. Therefore, the payoff to the arbitrageur at time t^* is positive. It remains to show that if $t^* \rightarrow \infty$, the payoff to the arbitrageur will also be positive.

Let $\bar{u}_t$ denote a process, implicitly defined by $e^{-rt} \bar{u}_t = u_0 + \int_0^t e^{-rs} (1 - r) F_s ds$, which provides an upper bound for all processes u_t satisfying Inequality (39). From the Equation, the deviation needs to decrease without bound to keep the arbitrageur from making a profit, which would violate Assumption 2.

Therefore, if $F_0 < \lambda S_0$, longing 1 futures and shorting λ spot is a random maturity arbitrage strategy. $\square$

Proof to Proposition 3:

Proof. The proof follows the same logic as Proposition 1. The only difference is we need to subtract a cost of C from the total payoff to the arbitrageur. We focus on the $t < r$ case and note that a similar argument would apply to the $t > r$ case. In the edge case where $t = r$, the same logic from Proposition 2 can be applied.

Scenario 1: If $F_0 > \lambda S_0 + C$, consider the first strategy in Table 1. Its payoff will be given by subtracting C from the RHS of Equation (34):

$$-e^{-rt}u_t + (u_0 - C) + (\kappa - \gamma - r) \int_0^t u_s e^{-rs} ds.$$

Following the same logic, to prevent the arbitrageur from making a profit, the lower bound process in Equation (36) becomes:

$$\underline{u}_t = e^{(\kappa - \gamma)t} (u_0 - C).$$

When $u_0 - C > 0$, this process will explode toward ∞ at an exponential rate. Following the same argument as in proof proposition 1, the proposed trading strategy is a random-maturity arbitrage strategy.

Scenario 2: If $F_0 < \lambda S_0 - C$, then we show the second strategy in Table 1 is a random maturity arbitrage strategy. Following the same logic as the proof in Proposition 1, the payoff in RHS of Equation (38) becomes:

$$e^{-rt}u_t - (u_0 + C) - \int_0^t (1 - r)F_s ds.$$

To prevent the arbitrageur from making a profit, the upper bound process defined from Inequality (10) becomes:

$$e^{-rt}u_t \leq u_0 + C + \int_0^t e^{-rs}(1 - r)F_s ds$$

When $u_0 + C < 0$, the upper bound process will decrease toward $-\infty$ without bound. Therefore, the proposed strategy is a random-maturity arbitrage strategy.

In sum, with trading cost, the absence of random maturity arbitrage opportunity implies that the futures price lies within a bound relative to the spot: $|F_t - \lambda S_t| \leq C$. $\square$